

# A.E.T.H.E.R — PROJECT BIBLE

## Honeywell Design-A-Thon RVCE 2026 · Problem DP7 — Autonomous Navigator for GPS-Denied Environments

Read this first, every working session. When a build decision is ambiguous, the principle that wins is: *the IMU is the spine; vision is an aiding source that bounds inertial drift; and the system must always know — and prove — how wrong it is, relative to where it started.*

The spine (say it verbatim, 3 times — open, demo, close):

*"When the camera dies, our drift bound still covers the true position 95%+ of the time — and we know within half a second."*

### ✓ MEASURED RESULTS (build complete — 10 Jun)

The design below was **built and measured**. Every number has a committed artifact.

| Claim                                                                       | Measured                                                                                                    | Provenance                                                                            |
|-----------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Real VIO drift — OpenVINS stereo-MSCKF on OUR 202.2 m Gazebo tunnel flight  | <b>0.219 %</b> terminal (DP7 < 1.5 % — beaten 6.8x) · RMS ATE 0.202 m · seg-drift 0.94 %@10 m → 0.31 %@80 m | <code>results/drift_report.txt</code> (rerun: <code>docker compose run chain</code> ) |
| Integrity bound honest on the <b>real</b> VIO output                        | covers the true error <b>97.2 %</b> of 31,173 live verdicts                                                 | <code>results/integrity_chain.csv</code>                                              |
| Live integrity rig (real ROS2, injected faults)                             | coverage <b>97.6 %</b> emergent · <b>ANEES 3.05</b> ( $\chi^2_3 \approx 3$ ) · detection <b>1.18 s</b>      | <code>results/metrics.csv</code> , <code>docs/figures/measured_*.png</code>           |
| Physics-real 3D flight (X3 multicopter, rotor forces, closed-loop guidance) | full 200 m corridor, centered $\pm 0.58$ m, IMU alive (cruise $\sigma$ 0.287 m/s <sup>2</sup> )             | <code>bags/tunnel_mini</code> (GT+IMU, committed)                                     |
| Workspace quality                                                           | CI green: lint + tests + full <code>ros:jazzy</code> colcon build every push                                | GitHub Actions                                                                        |

**Judge-facing demo scenes** (see `docs/JUDGES.md` , `docs/RUNBOOK.md` ): `./scripts/judge_demo.sh live3d` — Gazebo 3D patrol + RViz + clickable **Mission Control** fault rail (kill camera / IMU bias / starvation / UWB aid); `tour` — narrated fault pass; `chain` — the real OpenVINS number regenerated live; `replay` — the guaranteed no-Gazebo rig.

**Build deltas vs the plan below** (the plan called them risks; here is how they resolved): OpenVINS does not build on Ubuntu 26.04/Lyrical (CMake 4, Boost 1.90, removed `ament_target_dependencies` ) → the real-VIO chain ships in a pinned `ros:jazzy` container; open-loop velocity flight drifted into the tunnel walls twice → the flight director is closed-loop on simulator ground truth (the autopilot is not the system under test); OpenVINS marginalizes MSCKF features in bursts → the monitor gained a 1 s feature-window and source-profile configs ( `integrity.yaml` replay-rig scales / `integrity_ov.yaml` OpenVINS scales).

## 0. Meta

| <b>Project</b>                                    | A.E.T.H.E.R — Self-localizing INertial-spine navigator with Trust, INtegrity & Layered-resilience                                                          |
|---------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Event</b>                                      | Honeywell Design-A-Thon, RVCE 2026 (Honeywell Aerospace × RV College of Engineering)                                                                       |
| <b>Problem</b>                                    | DP7 — Autonomous Navigator for GPS-Denied Environments                                                                                                     |
| <b>Team</b>                                       | <b>Rayyan + Ashitha</b> (2 members; rules allow up to 4)                                                                                                   |
| <b>Judges / mentors</b>                           | Honeywell Aerospace navigation SMEs — <b>Faiz K.</b> & <b>Renju Chandrasekharapanicker</b>                                                                 |
| <b>Target depts</b>                               | CSE, ECE, IT                                                                                                                                               |
| <b>Key dates</b>                                  | Launch 1 Jun · Q&A 6 Jun · <b>Submission closes 8 Jun</b> · SME shortlisting 10 Jun · In-person 3-hr final 12 Jun                                          |
| <b>8-Jun deliverable</b>                          | Design + solution + "the journey" <b>presentation</b> (PPT) + idea brief. Optional prototype/GitHub link. (Full working build = path to the 12-Jun final.) |
| <b>Rubric (official "tips to a better score")</b> | Problem Understanding · Planning/Approach · Innovativeness · Presentation                                                                                  |
| <b>Prizes</b>                                     | Winner ₹75,000 + Student Research Project · Runner-up ₹50,000 · two Third ₹25,000                                                                          |

# PART I — THE PITCH & THE SCIENCE

## 1. The problem (DP7)

**Goal (verbatim):** Develop a Visual-Inertial Odometry (VIO) system that enables accurate self-localization of a *simulated* drone in a GNSS-denied indoor environment.

**Design considerations (verbatim):** 1. State estimation: VIO drift < **1.5% over 200 m**. 2. Environment: simulated indoor space (tunnel/warehouse) without GPS. 3. Framework: ROS2-based pipeline running Gazebo or AirSim.

**Deliverables (verbatim):** 1. **Simulation setup** — Gazebo/AirSim tunnel/warehouse + simulated drone with stereo camera + IMU. 2. **VIO pipeline** — a working ROS2 node performing real-time VIO on the simulated sensor data. 3. **Performance report** — comparative graphs: Estimated Trajectory vs Ground Truth (from the simulator). 4. **Source code** — documented ROS2 workspace incl. custom VIO node + launch files.

### 1.1 Decoding < 1.5% over 200 m (most teams compute this wrong)

- It is the **KITTI-style translational-drift metric**, *not* a globally-aligned ATE number:  $\text{drift\%} = \frac{\|p_{\text{est}}(\text{end}) - p_{\text{gt}}(\text{end})\|}{\text{segment\_length} \times 100}$ , averaged over many fixed-length segments.
- 1.5% over 200 m**  $\Rightarrow \leq 3.0$  m terminal error after a 200 m path.
- Indoor scale  $\Rightarrow$  use **short segments (10 / 20 / 40 / 80 m)** (KITTI uses 100–800 m outdoors). OpenVINS `ov_eval` natively cross-checks at 8/16/24/32/40/48 m.
- This is **RPE-style (relative/odometry error)** — and that distinction is the foundation of our whole integrity story: the spec measures *relative* error, and our integrity bound bounds *relative* error. Same datum by design.

**Honest framing:** the 1.5% bar is a **floor, not a goal**. Off-the-shelf stereo-inertial VIO already clears it in sim (OpenVINS  $\approx$  0.1–0.3% drift on EuRoC; VINS-Fusion  $\approx$  0.0115 m ATE; ORB-SLAM3  $\approx$  1.12% on outdoor KITTI). **~80–90% of teams will also hit it.** The win is owning the regime where that number quietly falls apart — and proving our error bound stays honest through it.

## 2. The thesis & the wedge

**Thesis (for the SMEs):** *A.E.T.H.E.R. is a ROS2 stereo-inertial navigator for a GPS-denied indoor drone that publishes a live, fault-hypothesis-derived relative integrity bound (the vision analogue of GNSS RAIM, on the odometry solution DP7 actually specifies), detects loss of observability before drift explodes, and degrades gracefully to inertial dead-reckoning whose error growth is measured against a Honeywell HG4930-class IMU model — an open-source, student-scale instantiation of Honeywell's HANA resilient-PNT architecture.*

**Positioning in one breath:** *"Most teams will run a VIO and plot a trajectory. We built the layer Honeywell ships — the one that knows when the navigation solution can no longer be trusted, proves its own error bound covers the truth, and fails safely with a known, characterized error."*

**The wedge (say out loud):** *"Raw VIO accuracy is table-stakes — 80–90% of teams will clear 1.5%/200 m. For a panel that built HANA and ships RAIM-class integrity, the differentiated value is trust: a navigation solution that knows when it can no longer be trusted and degrades gracefully. HANA's first release is vision-aided nav with INS fallback; our integrity bound is the RAIM analogue for that vision aid."*

### 2.1 Self-scoring rubric-delta table

| Rubric axis           | Median "VINS-Fusion flythrough" team      | A.E.T.H.E.R.                                                                                                                        |
|-----------------------|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| Problem Understanding | "We built a VIO, it localizes"            | DP7 = Assured PNT; 6 named indoor failure modes $\rightarrow$ a bound that protects each; honest RL $\rightarrow$ estimation pivot  |
| Planning / Approach   | one monolithic clone, one trajectory plot | modular certifiable ROS2 node graph (any subset runs); locked Day-2 floor + fenced upside                                           |
| Innovativeness        | a trajectory + an ATE number              | observability-conditioned, fault-hypothesis-derived, 3-class <b>relative integrity bound</b> + measured <code>R(D)</code> ablation  |
| Presentation          | a clean flythrough                        | kill-the-camera beat sheet: breathing ellipse chasing a moving truth dot, audible alarm, 3-grade IMU ghost ladder, spine $\times 3$ |

## 3. Why Honeywell (tailor everything to this panel)

Honeywell Aerospace is a world leader in inertial navigation and **Assured / Resilient PNT** for GPS-denied and contested ops. **DP7 is not academic — it is an active Honeywell product battleground**, and our demo is a microcosm of it.

- **HANA — Honeywell Alternative Navigation Architecture** (launched **Oct 2025**): a *software-based, multi-system, layered* platform that **enhances INS performance by fusing add-on modalities**. Initial release = **vision-aided navigation**; MagNav + LEO-satellite modes on the 2026 roadmap. *The judges literally built HANA*. A.E.T.H.E.R is pitched as **"an open-source, indoor, student-scale HANA."**
- **Honeywell's mental model**: *INS is the spine; everything else is an aiding source that bounds INS drift*. We **invert the naive student view** — the IMU is not a helper for the camera; the camera is an aiding source that bounds the IMU, and on vision loss we degrade to inertial-only with **known, characterized** error growth.
- **Prized attributes** → **mapped to our system**: multi-sensor **fusion** · **redundancy** · **graceful degradation** · navigation **integrity / health monitoring** (the INS analogue of GNSS RAIM) · **bounded / known error** · **safety-criticality** · **certification** (DO-178C / DO-254 / ARP4754A / ARP4761).
- **Proof points to cite for currency**: U.S. Army **APEX / White Sands (Sept 2025)** resilient-nav demo; **HANA anti-jam launch (Oct 2025)**; **MagNav** on an Embraer E170 at **25 m CEP50** in GPS-denied flight; **HGuide n580** INS/GNSS (HG4930 IMU + multi-GNSS) and Honeywell's own **HGuide-n580-with-ROS** point-cloud whitepaper (proof ROS integration is on their radar).
- **Product lineage to name correctly**: HGuide **n580** INS/GNSS · HGuide **o180 / i300** OEM IMUs · tactical **HG1700 / HG4930** IMUs · Vision-Aided Navigation (VAN) · MagNav · AH-2000 AHRS (ships integrity monitoring).

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## 4. The journey (the rules demand it — and it scores Problem Understanding)

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*"We first framed GPS-denied flight as a reinforcement-learning decision problem (PPO + a risk slider, a procedural sim, a live dashboard). Reading DP7 through a navigation lens, we realized it is fundamentally a bounded-error state-estimation + integrity problem — 'where am I, and how wrong could I be?' — not a 'where should I go?' policy problem, and re-architected around exactly that. We kept what transferred: the build speed, the procedural map (now a low-texture stress generator), the risk slider (now an operator integrity-threshold dial), and the dashboard (now the trust HUD). That system is A.E.T.H.E.R — Assured Estimation with Trust, Health & Error-bounded Reckoning: integrity-aware, observability-conditioned VIO in a HANA-style resilient stack."*

This re-framing — from a decision problem to a bounded-error estimation+integrity problem — is exactly what scores Problem Understanding with a navigation panel: it shows we found the right problem class.

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## 5. System architecture — the schema

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Three-layer **"Integrity-Aware VIO."** Each box is an **independently-runnable ROS2 node** — so *any subset runs* = graceful degradation is **proven by the node graph itself**, and module failures never cascade.



**Node ownership (2-person split):** - Rayyan — `vio_core` + `integrity_monitor` (Layer 1 + Layer 2: the OpenVINS fork, `/vio/internals` hook, the integrity math). - Ashitha — `sensor_bridge` + sim worlds + `degradation_manager` + `trajectory_publisher` /cockpit + evaluation/report/deck (Layer 3 + sim + eval + UI).

## 6. Technical deep-dive

### 6.1 Layer 1 — the VIO core

**OpenVINS ( `ov_msckf` ), tightly-coupled stereo-inertial MSCKF/EKF.** Chosen **primary** over VINS-Fusion / ORB-SLAM3 because it directly serves the integrity layer: - Exposes **clean state covariance + per-feature innovations** ( `v` , `S` , `H` ) — the raw material for NIS / NEES / the integrity bound. (Optimization back-ends bury this.) - **Constant-time / real-time** on a laptop CPU. - Built-in **zero-velocity update** and **First-Estimate Jacobians (FEJ)** for observability consistency. - Ships `ov_eval` (ATE + segment-RPE + NEES) and **launches on EuRoC out of the box** — the Day-1 de-risk path.

| Stage              | What we use                          | What we customize                                                                                                                                                        |
|--------------------|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Front-end          | OpenVINS KLT stereo tracker + RANSAC | Expose tracked-feature count, parallax, info matrix <code>Λ_v</code> to <code>integrity_monitor</code> ; ( <i>stretch</i> ) XFeat / SuperPoint+LightGlue for low-texture |
| IMU preintegration | Forster on-manifold preintegration   | Inertial-only path = degraded-mode dead-reckoning, driven by HG4930 + Gauss-Markov bias model                                                                            |
| Back-end           | MSCKF sliding-window EKF + FEJ       | Tap <code>P_xy</code> , per-feature <code>v_i</code> , <code>S_i</code> , null-space-projected <code>H_x,i</code> each update                                            |
| Tightly-coupled?   | <b>Yes</b> — joint raw-feature + IMU | Stated explicitly (field literacy + Honeywell's deep-fusion preference)                                                                                                  |
| Loop closure       | <b>Off</b> for headline numbers      | Loop closure masks the very drift we measure; noted as a path to a <i>global</i> bound (roadmap)                                                                         |

**Backup / comparison:** VINS-Fusion (HKUST, ROS2 Humble port) for one report row. **ORB-SLAM3** cited as the accuracy benchmark, not a dependency.

**The integration crux (de-risk Day 0):** the integrity layer needs per-feature internals OpenVINS does **not** publish by default. Hook location: `ov_msckf` 's `UpdaterMSCKF::update()` ( `ov_msckf/src/update/UpdaterMSCKF.cpp` ), after triangulation + left-null-space projection of `H_f` . Publish a custom `VioInternals.msg` = {`header`, `state_cov(P)`, `feature_id[]`, `nu[]`, `S[]`, `Hx_row[]`, `chi2[]`, `n_features`} on `/vio/internals` via a read-only shim (no estimator-logic change). **Wire + screenshot this tonight on both laptops** → "named the hook" becomes "showed the hook working." Fallback: dump internals to bag → run the integrity layer **offline** for all report graphs; a feature-count +  $\chi^2$  proxy drives the live trust light.

6.2 Layer 2 — the integrity layer (the marquee)

6.2.0 Relative to WHAT datum (read first). Pure VIO has no absolute reference — global position + yaw are unobservable, so only the **local/relative trajectory** is well-constrained. Therefore our integrity bound **IB** is a **relative (local-frame) bound on the odometry solution — the exact quantity DP7 specifies via segment-RPE** — *not* a global-position Protection Level. A global PL needs an absolute aid (UWB / loop closure) — that is our stretch and **roadmap delta #1**. This is *why*  $\sigma_{pos}$  is the **instantaneous filter uncertainty**, not accumulated ATE, and why the bound is evaluated the same way `evo_rpe --align_origin` evaluates the spec.

6.2.1 The observability index **D** (not a relabeled condition number). 1. Form the visual information matrix  $\Lambda_v = \sum_i \hat{H}_{x,i}^T \hat{R}_i^{-1} \hat{H}_{x,i}$  over active features. 2. **Whiten by covariance**:  $\tilde{\Lambda}_v = P^{1/2} \Lambda_v P^{1/2}$  → dimensionless **information-ratio** units (proof: variance × inverse-variance cancels). An eigenvalue = "how much info vision adds along this direction, relative to current uncertainty." 3. Eigendecompose each update; **flag** any eigenvector below **0.10×** the per-axis healthy-run spectral floor (calibrated offline on the texture-rich world). 4. **Identify the motion-degenerate subspace online** from the IMU-propagated velocity/angular-rate: classify regime (near-constant-velocity → metric scale unobservable; near-pure-rotation → no triangulation; near-planar → out-of-plane), form the known analytic degenerate directions, and **exclude the 4 OC-EKF structural nulls** (global position ×3 + yaw). 5.  $D \in [0,1]$  = normalized energy of flagged eigenvectors lying in that motion-degenerate subspace — *which* state direction vision can no longer constrain. Label which mode fired on the slide. - **Grounding**: Hesck & Kottas (OC-EKF VINS); Zhang & Singh, *On Degeneracy of Optimization-based State Estimation*, ICRA 2016 (the eigenvalue-floor method we whiten and adapt).

6.2.2 The association-fault integrity bound (the RAIM analogue):  $IB_{assoc} = k_{ffd} \cdot \sigma_{pos} + slope_{max} \cdot p_{bias}$  (a bound on relative/local horizontal error) -  $\sigma_{pos} = \sqrt{\lambda_{max}(P_{xy})}$  — instantaneous 1-σ horizontal position uncertainty. -  $k_{ffd}$  from a stated **integrity-risk allocation**  $P_{HMI} = 1 \times 10^{-5}/hr$  (DAL-C indoor-drone budget). At 20 Hz, per-sample risk  $= 1.39 \times 10^{-10} \rightarrow k_{ffd} = \sqrt{\chi^2_{2-1}(1-1.39 \times 10^{-10}, 2)} = 6.74$  (not an ad-hoc  $k=3$ ). - **Fault hypothesis**: one corrupted feature track injecting bias **b** (mis-association in a self-similar corridor). -  $slope_{max}$  = the **RAIM slope** — worst-case undetected-bias contribution to horizontal position per unit  $\sqrt{\text{non-centrality}}$ , max over fault direction (a generalized-eigenvalue problem; see Appendix A). -  $p_{bias}$  = minimum detectable bias from the **non-central  $\chi^2$** :  $dof=2, P_{FA}=1 \times 10^{-3} \rightarrow T_D = \chi^2_{2-1}(0.999, 2) = 13.82; P_{MD}=1 \times 10^{-3} \rightarrow \lambda_{md} = 45.0$ , so  $p_{bias} = \sqrt{45.0} \cdot \sigma_{meas} = 6.71 \cdot \sigma_{meas}$ .

6.2.3 THREE bounds for THREE fault classes (every RAIM kill-shot pre-empted):

| Fault class                       | Trigger                                 | Detector                                   | Bound                                                                                                                                      |
|-----------------------------------|-----------------------------------------|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| (a) Single association fault      | one mismatch in a self-similar corridor | per-feature NIS $\chi^2$ FDE               | $IB_{assoc}$ (excludes it; $slope_{max} \cdot p_{bias}$ bounds the worst undetected residual)                                              |
| (b) Degeneracy / whole-frame loss | bare wall, blackout, motion blur        | $D \rightarrow 1$                          | $IB_{deg} = k_{ffd} \cdot \sqrt{\lambda_{max}(P_{xy})}$ , $P$ propagated through the HG4930 model (no $p_{bias}$ — no measurement to test) |
| (c) Correlated multi-track fault  | many features mismatch consistently     | <b>frame-level (global) NIS</b> + $D$ rise | same $IB_{deg}$ ; the global NIS trips and we degrade rather than trust a coherent-but-wrong fix                                           |

6.2.4 Covariance honesty (NEES). Nominal: NEES vs ground truth shows the filter isn't over-confident (near the  $\chi^2$  band). Outage: no innovation to test → validate the **bloomed covariance directly against ground truth** (the outage-window coverage plot) — a *characterized* growth governed by the HG4930 model, not a free parameter.

6.2.5 The dual bound (why  $k=6.74$  isn't trivially over-met). A  $k=6.74$  2-D bound has exceedance probability  $1.39 \times 10^{-10}$  — essentially un-violable, so we report a **dual bound** on the same plot and name the integrity/availability trade: - **Operational alert bound**  $k = 2.45$  (95% 2-D containment) — the tight, useful bound the trust light acts on. **The load-bearing number** (reported **held-out**, §8.3). - **DAL-C integrity bound**  $k_{ffd} = 6.74$  — conservative, ~100% by construction.

6.2.6 Trust score `/nav/trust` (0–1). Monotone fuse of normalized feature count, innovation health (fraction passing NIS), and  $D$ . Green  $\geq 0.66$ , amber 0.33–0.66, red  $< 0.33$ .

6.3 Layer 3 — the inertial fallback (honest physics)

The Gazebo IMU is parameterized to the Honeywell HG4930 CA51 datasheet (and we say so):

| Parameter                    | HG4930 CA51  | Gazebo <code>&lt;imu&gt;</code> SDF field                     |
|------------------------------|--------------|---------------------------------------------------------------|
| Gyro in-run bias instability | 0.25 °/hr    | Gauss-Markov + <code>angular_velocity.*.bias_stddev</code>    |
| Gyro ARW                     | 0.04 °/√hr   | <code>angular_velocity.*.noise_density</code>                 |
| Accel VRW                    | 0.03 m/s/√hr | <code>linear_acceleration.*.noise_density</code>              |
| Accel bias instability       | ~0.05 mg     | Gauss-Markov + <code>linear_acceleration.*.bias_stddev</code> |

The number that makes the demo honest: at 1.5 m/s cruise, an **8 s** vision outage on a true HG4930-class IMU yields only  $\approx 0.4$  m horizontal error — dominated by velocity-estimate uncertainty at outage entry ( $\sim 0.05$  m/s × 8 s  $\approx 0.40$  m), *not* IMU noise (gyro-bias  $t^3$ , accel-bias  $t^2$  terms  $< 2$  cm over 8 s). **A tactical-grade inertial core keeps the dead-reckoning bound tight — the whole Honeywell point, quantified.**

Three-grade IMU ladder (fair engineering comparison):

| Grade                                   | Accel bias | Gyro bias instab. | 8 s outage error | Role                                |
|-----------------------------------------|------------|-------------------|------------------|-------------------------------------|
| HG4930 (tactical, our spine)            | ~0.05 mg   | 0.25 °/hr         | $\approx 0.4$ m  | the tight, certifiable bound        |
| Industrial MEMS (realistic aided-drone) | ~1.5 mg    | ~3 °/hr           | $\approx 1.2$ m  | what a serious integrator would fly |

| Grade                      | Accel bias | Gyro bias instab. | 8 s outage error | Role                         |
|----------------------------|------------|-------------------|------------------|------------------------------|
| Raw consumer (phone-class) | ~20 mg     | ~30 °/hr          | ≈ <b>6.3 m</b>   | the visceral off-frame ghost |

**HG4930 envelope vs outage duration:**  $4\text{ s} \rightarrow \approx 0.20\text{ m} \cdot 8\text{ s} \rightarrow \approx 0.40\text{ m} \cdot 12\text{ s} \rightarrow \approx 0.60\text{ m} \cdot \mathbf{20\text{ s (demo)}} \rightarrow \approx \mathbf{1.0\text{--}1.1\text{ m}}$ .

**The falsifiable cross-check (beats "you tuned your own IMU"):** run the inertial-only channel against sim ground truth and show the *measured* dead-reckoning error matches the **analytic first-principles envelope (0.20 / 0.40 / 0.60 / 1.0 m) to within ±20%**. "*Our dead-reckoning matches physics, independent of the filter.*"

**Honesty notes:** Gazebo's stock IMU is white-noise + random-walk; we approximate the 1/f bias-instability floor with a **first-order Gauss-Markov** bias process (overlay simulated Allan deviation vs datasheet) and **cap the showcase outage ≤ 20 s**. We reserve the word **"bounded"** strictly for the UWB-aided case (a true global anchor); inertial-only error is "*slow, known, short-duration-limited.*" ZUPT fires only during genuine zero-motion (landed / commanded stop), never claimed to cap in-flight drift.

#### 6.4 How we achieve AND prove < 1.5% over 200 m

**Achieve:** (1) nominal margin — tightly-coupled stereo-inertial OpenVINS ≈ 0.1–0.3% drift in sim; (2) hard-stretch survival — in the feature-poor stretch the **R(D)** scheduler hands the solution to the IMU so error grows slowly/predictably (≈ 0.4 m/8 s) instead of the camera injecting garbage; re-acquisition re-converges. **Prove:** report drift the KITTI/RPE way ( `evo_rpe --delta 10 --delta_unit m --pose_relation trans_part` , error ÷ segment, averaged over 10/20/40/80 m; cross-check `ov_eval` ); use `--align_origin` (not full Umeyama) for honest accumulated drift; publish the **IB -vs- actual-error coverage** (dual bound) and **NEES** plots.

## 9. The demo (the 60-second gasp)

**Screen:** a full-pane **drone-eye camera view** (flies the curved corridor, then goes black); the breathing **IB ellipse + ghost traces large & center, chasing a moving truth dot**; a live coverage counter + green/amber/red trust light.

| t            | Beat                                   | On screen                                                                                                                                   | Callout                                                       |
|--------------|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|
| 0–8 s        | Nominal                                | flies @1.5 m/s; estimate tracks the moving GT dot; trust green; drift < 1.5%; small <b>IB</b> ellipse covering truth                        | calm                                                          |
| 8 s          | Inject fault (keypress)                | pane goes <b>black</b> ; stock-VIO ghost silently diverges                                                                                  | audible alarm + "STOCK VIO IS NOW LYING"                      |
| 8–9 s        | The system <i>knows</i>                | <b>D&gt;1</b> + global NIS trips → trust amber→red in <1 m / <10 frames → fallback to inertial; mode labelled ("scale unobservable")        | "INTEGRITY ALERT — DEGRADE TO INERTIAL"                       |
| 9–28 s       | Breathing ellipse chasing moving truth | <b>IB</b> blooms along the HG4930 curve to ~1.0–1.1 m, visibly chasing-and-containing the moving true dot; "TRUTH INSIDE BOUND: 100%" ticks | "The bound is breathing — and it never lets the truth out."   |
| same instant | Three-grade IMU ladder                 | tactical bound tight; industrial ghost ~1.2 m; consumer ghost shoots off-frame ~6.3 m                                                       | "Same outage, three IMU grades — 0.4 / 1.2 / off the screen." |
| 28–32 s      | Recovery                               | vision returns → filter re-converges; ellipse + trust snap to green                                                                         | "Vision back. Re-converged. Trust restored."                  |

**Demo ladder (rehearsed to muscle memory):** **A — Live** (deterministic launch, keypress fault; go/no-go: 5/5 clean rehearsals by Day-4 EOD or ship B) → **B — Rosbag replay** (same nodes, pre-recorded sim bag; integrity layer still runs live) → **C — Backup video** (pre-rendered capture). A scripted, deliberately-triggered red flash makes a transient red always read as *the feature working*, never an accident.

**Stretch 30-s UWB clip:** `robot_localization` EKF + one simulated UWB anchor → degraded-mode error goes from slowly-growing (vision-only) to **flat-bounded** (UWB-aided), **IB** collapsing to a globally-bounded ellipse. Labelled proof-of-concept of HANA's layered-modality philosophy (roadmap delta #1).

## 8. Feasibility re-scope — what we actually build vs what we present

This section is the honest engineering contract behind the pitch. It exists because "fully working on the 12th" must be a *guarantee*, not a hope. The system is tiered exactly like the product it describes: a locked floor, a de-risked live demo, and a clearly-fenced stretch — so whatever we finish is still a complete, winning DP7 submission.

**The timeline that governs everything.** Build window = **8, 9, 10, 11 June**; the 12th is the 3-hour in-person demo, not build time. So every "done" below means *done by end of 11 June*, by **two** second-year students (Rayyan + Ashitha) leaning heavily on AI-assisted development.

**The #1 risk is the environment, and it is Day-0.** ROS2 Humble + Gazebo Harmonic + PX4 want **Ubuntu 22.04**. Our dev machines are Windows 11. Native dual-boot is strongly preferred (best GPU + timing); a cloud GPU box is the fallback; WSL2 is a last resort and only viable for the integrity node

(which needs a rosbag, not live Gazebo rendering). Lose Day 1 to environment hell and the plan compresses — which is why it is the first thing we do, with a tested fallback chain (see §9).

The three tiers:

| Tier                         | What                                                                                                                                                                                                                                                                                                                                                                                                      | Confidence       | Detail                                                                                                                                                   |
|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| GUARANTEED FLOOR             | OpenVINS on the EuRoC dataset (out-of-box) <b>and</b> on a Gazebo Harmonic tunnel → drift < 1.5% → estimate-vs-ground-truth plots → clean documented ROS2 workspace = <b>ALL FOUR DP7 deliverables</b> .                                                                                                                                                                                                  | High (by 11 Jun) | Stock tools, mature open source. \$11 day-by-day locks this by end of Day 2.                                                                             |
| LIVE DEMO (de-risked)        | The <code>integrity_monitor</code> written in <b>Python</b> , consuming only what OpenVINS already publishes — pose+covariance on <code>/ov_msckf/odomimu</code> + tracked-feature count → a covariance-trace / protection-level <b>proxy</b> (the breathing bound), <b>NEES</b> vs ground truth, and the kill-camera → degrade-to-inertial → bound-blooms beat. <b>No C++ fork on the critical path.</b> | Medium-high      | \$10 details the node. The breathing-ellipse demo runs from this.                                                                                        |
| STRETCH (presented, not bet) | Forking OpenVINS C++ ( <code>UpdaterMSCKF::update</code> ) to publish per-feature internals; the full whitened-eigendecomposition observability index <code>D</code> ; the generalized-eigenvalue RAIM-slope per-feature 3-class NIS bound.                                                                                                                                                               | Day-4 only       | Presented as <b>derived + computed offline</b> on a dumped rosbag — a backup slide + one offline plot. <b>"Say the full math; run the Python proxy."</b> |

**The safety net.** The integrity demo can run on **EuRoC real data independently of Gazebo**, so even if sim VIO is shaky the demo still works and the Gazebo deliverable is satisfied with stock VIO + estimate-vs-GT plots. The worst realistic outcome is "less impressive," never "nothing to show."

**Honesty as a design element.** The 8-June submission is a **design proposal**; every trajectory / drift / coverage figure in the deck is stamped **"ILLUSTRATIVE / TARGET — not measured."** Measured numbers (EuRoC drift, the `R(D)` handover spike, NEES coverage) land during the build and replace the targets by the 12th.

## PART II — BUILD & OPERATIONS

### 9. Environment & Toolchain Setup (the #1 risk, de-risked)

The single largest schedule risk in this project is not the math — it is the operating system. ROS2 Humble, Gazebo Harmonic, and PX4 SITL are all first-class on **Ubuntu 22.04 LTS** and second-class everywhere else. Both team machines run Windows 11. **Day 0 (8 Jun) is therefore an OS day, not a code day.** If we lose Day 1 fighting the toolchain, the whole 4-day window wobbles. This section is the ordered, concrete plan to make that not happen, with a hard go/no-go gate at the end of Day 0.

Why Ubuntu 22.04 specifically (not 24.04, not Windows)

| Component                              | Required                       | Reason                                                                                                                                                                                                                                                                             |
|----------------------------------------|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ROS2 Humble Hawksbill                  | Ubuntu <b>22.04</b> (Jammy)    | Humble is the Tier-1 LTS for 22.04. ROS2 Jazzy targets 24.04 but the <code>ros-humble-ros-gzharmonic</code> bridge and OpenVINS build recipes are battle-tested on Humble/Jammy. Do <b>not</b> drift to 24.04 — you trade a known-good stack for an untested one with 3 days left. |
| Gazebo Harmonic (gz-sim8)              | Ubuntu 22.04 via OSRF apt repo | Harmonic is the LTS Gazebo paired with Humble through <code>ros_gz</code> . Installs cleanly on Jammy from <code>packages.osrfoundation.org</code> .                                                                                                                               |
| <code>ros-humble-ros-gzharmonic</code> | Ubuntu 22.04                   | The Humble→Harmonic bridge metapackage. This exact pairing is the one with apt binaries — don't improvise a Fortress/Garden mix.                                                                                                                                                   |
| PX4 SITL                               | Ubuntu 22.04                   | <code>Tools/setup/ubuntu.sh</code> from the PX4 tree targets 22.04. PX4-ROS2-Gazebo templates (SathanBERNARD, XTDrone) assume it.                                                                                                                                                  |
| OpenVINS                               | Any Linux w/ ROS2              | Builds against ROS2 + Ceres/OpenCV/Eigen/Boost; trivially happy on Jammy.                                                                                                                                                                                                          |

Decision: native dual-boot vs WSL2 vs cloud GPU

| Option                                                  | Pros                                                                                                                              | Cons                                                                                                                                                                                                                                                  | Verdict                                                                                                                                                        |
|---------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Native dual-boot</b> (Ubuntu 22.04 alongside Win 11) | Full GPU access; real-time timing for Gazebo physics + RViz; no virtualization jitter; the EXACT environment all the docs assume. | ~45 min one-time install; need ~80 GB free + a USB stick; partition anxiety.                                                                                                                                                                          | <b>RECOMMENDED — do this on at least one machine (Rayyan's, the demo machine).</b>                                                                             |
| <b>WSL2 (WSLg)</b>                                      | Stays inside Windows; fast to start.                                                                                              | GPU passthrough to Gazebo is flaky; OpenGL/Vulkan + WSLg compositing causes Gazebo Harmonic render stalls and clock-skew; RViz/ <code>gz sim</code> GUI timing is unreliable. Great for <i>headless</i> OpenVINS-on-EuRoC, poor for <i>live sim</i> . | <b>Backup / second machine only.</b> Acceptable for headless dataset runs and Python <code>integrity_monitor</code> dev; not trusted for the live Gazebo demo. |



| Option                                        | Pros                               | Cons                                                                                                | Verdict                                                                                                |
|-----------------------------------------------|------------------------------------|-----------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| <b>Cloud GPU box</b><br>(e.g. a 22.04 GPU VM) | Clean 22.04; strong GPU; nukeable. | Network latency to RViz/GUI; cost; rosbag upload friction; demo-day Wi-Fi dependency = a live risk. | <b>Last-resort fallback</b> if both laptops fail to dual-boot. Pre-bake an image Day 0 if used at all. |

**Plan of record:** dual-boot Ubuntu 22.04 on the demo machine; mirror onto the second machine, or run WSL2 there for parallel Python work. The demo on 12 Jun runs from the native dual-boot.

## Day 0 (8 Jun) ordered setup — copy-paste runbook

**Step 1 — Dual-boot Ubuntu 22.04 LTS.** Flash 22.04.x desktop ISO to USB (Rufus/balenaEtcher), shrink the Windows partition (~80 GB), install alongside Windows (keep both bootloaders). Verify GPU after first boot: `nvidia-smi` (if NVIDIA) or `glxinfo | grep "OpenGL renderer"`.

### Step 2 — ROS2 Humble (desktop).

```
sudo apt update && sudo apt install -y software-properties-common curl
sudo add-apt-repository universe -y
sudo curl -sSL https://raw.githubusercontent.com/ros/rosdistro/master/ros.key \
  -o /usr/share/keyrings/ros-archive-keyring.gpg
echo "deb [arch=$(dpkg --print-architecture) signed-by=/usr/share/keyrings/ros-archive-keyring.gpg] \
  http://packages.ros.org/ros2/ubuntu $(. /etc/os-release && echo $UBUNTU_CODENAME) main" \
  | sudo tee /etc/apt/sources.list.d/ros2.list > /dev/null
sudo apt update && sudo apt install -y ros-humble-desktop ros-dev-tools
echo "source /opt/ros/humble/setup.bash" >> ~/.bashrc && source ~/.bashrc
ros2 doctor    # sanity
```

### Step 3 — Gazebo Harmonic + ROS2 bridge.

```
sudo curl https://packages.osrfoundation.org/gazebo.gpg --output \
  /usr/share/keyrings/pkgs-osrf-archive-keyring.gpg
echo "deb [arch=$(dpkg --print-architecture) signed-by=/usr/share/keyrings/pkgs-osrf-archive-keyring.gpg] \
  http://packages.osrfoundation.org/gazebo/ubuntu-stable $(lsb_release -cs) main" \
  | sudo tee /etc/apt/sources.list.d/gazebo-stable.list > /dev/null
sudo apt update && sudo apt install -y gz-harmonic ros-humble-ros-gzharmonic
gz sim --version    # expect Gazebo Sim 8.x (Harmonic)
```

### Step 4 — PX4 SITL.

```
git clone https://github.com/PX4/PX4-Autopilot.git --recursive
bash ./PX4-Autopilot/Tools/setup/ubuntu.sh    # installs sim deps; reboot after
# smoke: make px4_sitl_gz_x500    (launches the default x500 quad in Gazebo)
```

**Step 5 — OpenVINS deps + build from source.** OpenVINS needs **Eigen3**, **Boost**, **OpenCV (≥4)**, and **Ceres Solver**. Install deps, then build in a clean colcon workspace:

```
sudo apt install -y libeigen3-dev libboost-all-dev libopencv-dev \
  libceres-dev python3-colcon-common-extensions
mkdir -p ~/ws_ov/src && cd ~/ws_ov/src
git clone https://github.com/rpng/open_vins.git
cd ~/ws_ov
colcon build --packages-select ov_core ov_init ov_msckf ov_eval \
  --cmake-args -DCMAKE_BUILD_TYPE=Release
source ~/ws_ov/install/setup.bash
```

Note: building **Release** matters — a Debug build of `ov_msckf` will not hit real-time and will muddy every latency number.

**Step 6 — Run OpenVINS on EuRoC out-of-box (the floor test).** Download a EuRoC MAV sequence (`V1_01_easy` is the gentlest first target; `MH_01_easy` next), convert/play as a ROS2 bag, and launch the stereo-inertial MSCKF with the stock EuRoC config:



```
# terminal A
ros2 launch ov_msckf subscribe.launch.py config:=euroc_mav
# terminal B (play the EuRoC sequence as a ROS2 bag)
ros2 bag play V1_01_easy
# OpenVINS publishes pose+covariance on /ov_msckf/odomimu (this is the topic
# the integrity_monitor will consume – confirm it ticks here)
ros2 topic hz /ov_msckf/odomimu
```

Then quantify against ground truth with **ov\_eval / evo**:

```
evo_ape tum groundtruth.txt stamped_traj_estimate.txt --align_origin # ATE
evo_rpe tum groundtruth.txt stamped_traj_estimate.txt # segment-RPE / drift %
```

Day 0 smoke-test: the go/no-go gate

Run these four checks **before close of 8 Jun**. All four green = proceed to Day 1 (Gazebo tunnel world + integrity\_monitor). Any red = stop and execute the matching fallback before doing anything else.

| # | Check                                                                                  | Pass criterion                                                                                          | Fallback if red                                                                                                                                                                      |
|---|----------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | <code>ros2 doctor</code> + <code>gz sim --version</code>                               | ROS2 Humble sourced; Gazebo Sim 8.x reports                                                             | Re-add apt repos; if Gazebo render fails under WSL2, that's expected → move sim work to the dual-boot machine.                                                                       |
| 2 | <code>make px4_sitl gz_x500</code>                                                     | x500 quad spawns and arms in Gazebo Harmonic                                                            | If PX4→gz bridge is broken, <b>defer PX4</b> — it's only needed for piloted flight; OpenVINS-on-EuRoC and rosbag replay don't require it. PX4 becomes a Day-3 stretch.               |
| 3 | OpenVINS on <b>EuRoC</b> runs; <code>/ov_msckf/odomimu</code> publishes at sensor rate | <code>ros2 topic hz</code> shows steady output; trajectory looks sane in RViz                           | If colcon build fails, it's almost always a Ceres/OpenCV version mismatch → pin <code>libceres-dev</code> / <code>libopencv-dev</code> from apt (not source), clean-rebuild Release. |
| 4 | <code>evo_rpe</code> on the EuRoC run                                                  | drift well under the <b>1.5% / ≤3.0 m over 200 m</b> DP7 bar (EuRoC out-of-box clears this comfortably) | If numbers are off, suspect the GT alignment, not the VIO → re-run with <code>--align_origin</code> and confirm TUM timestamp matching.                                              |

**Why this gate de-risks the whole build:** passing checks 3 and 4 means the **guaranteed floor** — OpenVINS + estimate-vs-GT + a documented colcon workspace — is already in hand on Day 0 using stock software and a public dataset. That alone satisfies the spirit of all four DP7 deliverables. Everything after Day 0 (the Gazebo tunnel world, and the Python `integrity_monitor` consuming `/ov_msckf/odomimu` pose-covariance + feature count) is **additive**, layered on a base we've already proven. The live demo never depends on a C++ fork, and the integrity layer can fall back to EuRoC real data if the Gazebo sim is shaky — so a red on check 2 (PX4) or a wobbly Gazebo world cannot sink the project.

**Honesty note for the 8-Jun deck:** any trajectory or drift chart in the design deck is **illustrative / target**, not a measured result — label it as such. The first *measured* numbers land at Day-0 smoke-test check 4 and get reported for real in the performance report.

10. The Python `integrity_monitor` Architecture (De-Risked Live Demo)

This is the node that earns the "innovativeness" points without betting the demo on a C++ fork. It is a **pure-Python `rclpy` node** that subscribes only to topics OpenVINS *already publishes out of the box* and turns them into a live, fault-bounded RELATIVE integrity layer — the RAIM-for-vision analogue on the segment-RPE quantity DP7 specifies. No fork of `ov_msckf` is required for anything you see on screen on 12 Jun.

The design rule is the same one that governs the whole stack: **say the full math, run the Python proxy**. Everything the judges *see move* is computed in this node from published covariance and feature counts. Everything that needs per-feature MSCKF internals is presented as DERIVED math on a slide and, at most, computed offline once on a dumped rosbag.

Topic contract

The node sits strictly downstream of `vio_core` and upstream of `degradation_manager` / `health_cockpit`. It consumes:

| Topic                                                                          | Type                                 | Source                                                         | Used for                                                                                                   |
|--------------------------------------------------------------------------------|--------------------------------------|----------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| <code>/ov_msckf/odomimu</code>                                                 | <code>nav_msgs/Odometry</code>       | OpenVINS <code>ov_msckf</code>                                 | IMU-rate pose <b>with the 6×6 pose covariance</b> in <code>pose.covariance</code> — the spine of the proxy |
| <code>/ov_msckf/trackhist</code> (or <code>/ov_msckf/points_msckf</code> size) | <code>sensor_msgs/PointCloud2</code> | OpenVINS                                                       | tracked-feature <b>count</b> N_feat (vision-health signal)                                                 |
| <code>/sentinel/ground_truth</code>                                            | <code>nav_msgs/Odometry</code>       | Gazebo <code>ros_gz_bridge</code> (PX4 SITL ground-truth pose) | NEES vs truth (sim only)                                                                                   |

OpenVINS publishes `pose.covariance` in `odomimu` as the marginal covariance of the IMU pose state (orientation block + position block). We read the **3×3 position sub-block** `P_pp` (indices 21,22,23,27,28,29,33,34,35 of the row-major 6×6 — the lower-right position partition) and, separately, the full 6×6 trace for the scalar trust signal.

It publishes:

| Topic                             | Type                                      | Rate                    | Payload                                                                                        |
|-----------------------------------|-------------------------------------------|-------------------------|------------------------------------------------------------------------------------------------|
| <code>/nav/trust</code>           | <code>std_msgs/Float32</code>             | 20 Hz                   | scalar 0–1 trust (1 = nominal, 0 = blind)                                                      |
| <code>/nav/integrity_bound</code> | <code>geometry_msgs/Vector3Stamped</code> | 20 Hz                   | per-axis relative protection level (m) → the breathing-ellipse semi-axes                       |
| <code>/nav/state</code>           | <code>std_msgs/String</code>              | 20 Hz + on-change latch | <code>NOMINAL</code> / <code>DEGRADED</code> / <code>INERTIAL</code> / <code>RE-ACQUIRE</code> |
| <code>/nav/nees</code>            | <code>std_msgs/Float32</code>             | 20 Hz                   | normalized estimation error squared (consistency check, sim only)                              |

Covariance-trace → protection-level PROXY

The honest framing: a true GNSS-style Protection Level is a fault-hypothesis-derived bound. We compute a **filter-covariance-derived proxy of that bound** on the position state OpenVINS reports, scaled by a sigma multiplier `k`. This is the breathing ellipse.

For the position sub-block `P_pp` (3×3), the per-axis 1σ uncertainties are the square-roots of the diagonal, and the protection-level proxy is:

```
PL_axis[i] = k * sqrt(P_pp[i,i])      for i in {x, y, z}
```

We expose the **dual bound** so the slide and the demo agree:

| Bound         | <code>k</code> | Meaning                                                   | Provenance                                |
|---------------|----------------|-----------------------------------------------------------|-------------------------------------------|
| Operational   | <b>2.45</b>    | 95% containment (2D Rayleigh-ish operating point)         | what the breathing ellipse uses live      |
| DAL-C "as-if" | <b>6.74</b>    | DO-178C-style integrity allocation, $\lambda_{md} = 45.0$ | <code>scipy.stats</code> verified offline |

The DAL-C `k = 6.74` comes from the missed-detection allocation `λ_md = 45.0`; with the proxy at the EuRoC operating point this gives **IB ≈ 1.31 m covering a true error ≈ 0.3 m** — i.e. the bound is conservative by design (it *over-covers*, which is the correct failure direction for an integrity bound). These numbers are `scipy.stats`-reproducible and live in `bounds.py` so the deck and the node never disagree.

The ellipse the cockpit draws is `(PL_x, PL_y)` as semi-axes, centred on the OpenVINS estimate. "Breathing" = `P_pp` grows monotonically the instant vision stops updating the filter, so the ellipse blooms on its own — no scripting.

NEES vs ground truth (consistency proof, sim only)

NEES is what lets us *claim* the bound bounds, rather than assert it. With estimate `x̂`, truth `x` (from `/sentinel/ground_truth`), and position covariance `P_pp`:

```
e = x_pos - x_pos      # 3-vector
NEES = e^T * inv(P_pp) * e      # scalar, chi-square dim 3
```

A consistent filter has `E[NEES] ≈ 3` (the state dimension). We publish instantaneous NEES on `/nav/nees` and accumulate the average; the **acceptance gate is the two-sided 95% chi-square interval for dim 3 over N samples** (`scipy.stats.chi2.interval(0.95, 3)` → roughly [0.35, 7.81] single-sample; tightened by N for the time-averaged ANEES). If ANEES sits inside the band, the covariance — and therefore the protection-level proxy built from it — is statistically trustworthy. This is the one plot that converts "nice ellipse" into "RAIM-grade claim" in front of navigation SMEs.

Degradation state machine

The state is a continuous function of two observed quantities: vision health `N_feat` (tracked features) and a covariance-growth rate `tr = d/dt tr(P_pp)`. Trust is the smooth signal; the discrete state is the latched, hysteretic version the demo narrates.

```
trust = clamp( w1 * f(N_feat) + w2 * g(tr(P_pp)) , 0, 1 )
f(N_feat) = N_feat / N_nominal      (feature richness, capped at 1)
g(tr)     = tr_nominal / tr         (tighter covariance = healthier)
```

| State           | Entry trigger                                                  | Trust band | Behaviour                                                | Demo beat               |
|-----------------|----------------------------------------------------------------|------------|----------------------------------------------------------|-------------------------|
| <b>NOMINAL</b>  | <code>N_feat ≥ 80</code> and <code>tr = 0</code>               | ≥ 0.8      | full stereo-inertial, ellipse small & steady             | "tracking, bound tight" |
| <b>DEGRADED</b> | <code>N_feat</code> drops below 80 or <code>tr</code> positive | 0.4–0.8    | vision still aiding but thinning; ellipse starts growing | camera occluding        |

| State      | Entry trigger                                               | Trust band     | Behaviour                                                | Demo beat                                    |
|------------|-------------------------------------------------------------|----------------|----------------------------------------------------------|----------------------------------------------|
| INERTIAL   | <code>N_feat &lt; 10</code> for $\geq 0.3$ s (camera black) | < 0.4          | IMU-only dead-reckon; ellipse blooms along weak axis     | <b>camera kill → trust RED in &lt; 0.5 s</b> |
| RE-ACQUIRE | <code>N_feat</code> recovers $\geq 40$ after INERTIAL       | rising 0.4→0.8 | vision re-locking; ellipse contracts back over truth dot | camera restored, bound shrinks               |

Transitions use **hysteresis + a debounce timer** (enter INERTIAL only after `N_feat < 10` persists  $\geq 0.3$  s; this both kills chatter and is exactly the "we know within half a second" claim — the detection latency is a *designed* number, not an accident). The `< 0.5 s` to RED is the debounce window ( $\leq 0.3$  s) plus one or two 20 Hz publish ticks.

### What is LIVE-Python vs STRETCH-C++/OFFLINE-DERIVED

This table is the de-risking, stated plainly so the judges (and we) know exactly what is running:

| Capability                                                                       | Status                    | Where it runs                                      | Notes                                                                     |
|----------------------------------------------------------------------------------|---------------------------|----------------------------------------------------|---------------------------------------------------------------------------|
| Pose+covariance subscribe, trace proxy <code>k.vdiag</code>                      | LIVE Python               | <code>integrity_monitor</code> node                | from <code>/ov_msckf/odomimu</code> , no fork                             |
| Breathing integrity ellipse (semi-axes = <code>PL_x</code> , <code>PL_y</code> ) | LIVE Python               | node → <code>/nav/integrity_bound</code> → cockpit | grows from real <code>P_pp</code> , unscripted                            |
| <code>/nav/trust</code> , <code>/nav/state</code> machine + 0.3 s debounce       | LIVE Python               | node                                               | the kill-camera beat sheet                                                |
| NEES / ANEES vs sim ground truth + chi-square gate                               | LIVE Python               | node + <code>scipy.stats</code>                    | proves the bound bounds                                                   |
| Dual bound ( $k = 2.45 / 6.74$ , $\lambda_{md} = 45.0$ )                         | LIVE Python               | <code>bounds.py</code>                             | scipy-verified constants                                                  |
| Per-feature NIS, 3-class accept/suspect/reject                                   | STRETCH / OFFLINE-DERIVED | fork of <code>UpdaterMSCKF::update</code>          | math on slide; one offline plot on a dumped rosbag                        |
| Whitened-eigendecomposition observability index <b>D</b>                         | STRETCH / OFFLINE-DERIVED | post-processed <code>P</code> dump                 | "knows which direction vision can't see" — Zhang&Singh / OC-EKF grounding |
| Generalized-eigenvalue RAIM-slope per feature                                    | STRETCH / OFFLINE-DERIVED | offline notebook                                   | the full GNSS-RAIM-slope analogue                                         |

The split is deliberate: the **left column is everything the live demo depends on and is buildable in the 8–11 Jun window from stock OpenVINS output**; the right column is the full Honeywell-grade math, said out loud and shown as derived/offline, never bet on for the live run.

## Minimal node skeleton (pseudocode)

```
import rclpy
from rclpy.node import Node
from nav_msgs.msg import Odometry
from sensor_msgs.msg import PointCloud2
from std_msgs.msg import Float32, String
from geometry_msgs.msg import Vector3Stamped
import numpy as np
from scipy.stats import chi2
from bounds import K_OP, K_DALC          # 2.45, 6.74 (scipy-verified,  $\lambda_{md}=45.0$ )

N_NOMINAL, TR_NOMINAL = 120.0, 0.02
DEBOUNCE_S = 0.3

class IntegrityMonitor(Node):
    def __init__(self):
        super().__init__('integrity_monitor')
        self.create_subscription(Odometry, '/ov_msckf/odomimu', self.on_odom, 20)
        self.create_subscription(PointCloud2, '/ov_msckf/points_msckf', self.on_feat, 10)
        self.create_subscription(Odometry, '/sentinel/ground_truth', self.on_gt, 20)
        self.pub_trust = self.create_publisher(Float32, '/nav/trust', 10)
        self.pub_bound = self.create_publisher(Vector3Stamped, '/nav/integrity_bound', 10)
        self.pub_state = self.create_publisher(String, '/nav/state', 10)
        self.pub_nees = self.create_publisher(Float32, '/nav/nees', 10)
        self.create_timer(0.05, self.tick)          # 20 Hz
        self.P_pp = np.eye(3) * 1e-3                # position cov sub-block
        self.x_est = self.x_gt = None
        self.n_feat = N_NOMINAL
        self.tr_prev = None; self.tr_rate = 0.0
        self.state = 'NOMINAL'; self.t_below = None

    def on_odom(self, msg):
        C = np.array(msg.pose.covariance).reshape(6, 6)
        self.P_pp = C[3:6, 3:6]                    # lower-right = position block
        self.x_est = np.array([msg.pose.pose.position.x,
                               msg.pose.pose.position.y,
                               msg.pose.pose.position.z])

    def on_feat(self, msg): self.n_feat = msg.width # tracked-feature count
    def on_gt(self, msg):   self.x_gt = np.array([msg.pose.pose.position.x,
                                                  msg.pose.pose.position.y,
                                                  msg.pose.pose.position.z])

    def tick(self):
        # --- protection-level proxy (breathing ellipse) ---
        sigma = np.sqrt(np.clip(np.diag(self.P_pp), 1e-9, None))
        pl = K_OP * sigma                          # operational k=2.45
        v = Vector3Stamped(); v.vector.x, v.vector.y, v.vector.z = map(float, pl)
        self.pub_bound.publish(v)

        # --- covariance-growth rate + trust ---
        tr = float(np.trace(self.P_pp))
        if self.tr_prev is not None:
            self.tr_rate = (tr - self.tr_prev) / 0.05
        self.tr_prev = tr
        trust = float(np.clip(0.6*min(self.n_feat/N_NOMINAL, 1.0)
                              + 0.4*min(TR_NOMINAL/max(tr,1e-9), 1.0), 0, 1))
        self.pub_trust.publish(Float32(data=trust))

        # --- NEES vs ground truth (sim) ---
        if self.x_est is not None and self.x_gt is not None:
            e = self.x_est - self.x_gt
            nees = float(e @ np.linalg.inv(self.P_pp) @ e)
            self.pub_nees.publish(Float32(data=nees)) # gate: chi2.interval(0.95, 3)
```

```
# --- latched, debounced state machine ---
now = self.get_clock().now().nanoseconds * 1e-9
if self.n_feat < 10:
    self.t_below = self.t_below or now
    if now - self.t_below >= DEBOUNCE_S: self.state = 'INERTIAL'
else:
    if self.state == 'INERTIAL' and self.n_feat >= 40: self.state = 'RE-ACQUIRE'
    elif self.n_feat < 80 or self.tr_rate > 0:         self.state = 'DEGRADED'
    else:                                             self.state = 'NOMINAL'
    self.t_below = None
self.pub_state.publish(String(data=self.state))
```

The whole node is ~200 lines, depends on `rcipy`, `numpy`, `scipy`, and *nothing inside OpenVINS' C++*. It is the smallest possible artifact that lets us stand in front of Honeywell navigation SMEs and say the spine line — **"when the camera dies, our drift bound still covers the true position 95%+ of the time, and we know within half a second"** — and then show it happening live.

## 11. Guaranteed-Build Day-by-Day (8–11 June, 2 People)

**The contract:** by **end of Day 2 (9 Jun)** all four DP7 deliverables are LOCKED (the *Floor*); Days 3–4 are a *fenced Upside lane* that can be abandoned at any gate without touching the Floor. Two owners, parallel tracks:

- **Rayyan** → `vio_core` + `integrity_monitor` (OpenVINS bring-up, the Python integrity proxy).
- **Ashitha** → `sim/sensors` + `degradation_manager` + `eval/deck` (Gazebo world, sensor\_bridge, evo/ov\_eval, the deck + report).

Convention: each block ends with a **commit to the shared `sentinel_ws` repo**. Daily 19:00 **15-min sync + gate check**.

### Day 0 / pre-8th — environment is the real Day 1 risk

**Do this the night of the 7th if at all possible.** WINDOWS→UBUNTU is the #1 risk. Native dual-boot Ubuntu 22.04 strongly preferred; WSL2/VM Gazebo is fragile for GPU/timing; a cloud GPU box (e.g. a spot Ubuntu 22.04 instance) is the fallback. **If both laptops are still on Windows by 8 Jun 10:00, that is a RED gate — both people stop and fix the OS before anything else.**

### Day 1 — 8 Jun (design deck due today + EuRoC floor)

| Time                | Rayyan ( <code>vio_core</code> / <code>integrity</code> )                                                                                                                   | Ashitha ( <code>sim</code> / <code>eval</code> / <code>deck</code> )                                                                             |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| 09:00–11:00         | Install <b>ROS2 Humble</b> + colcon; <code>git clone</code> OpenVINS, build <code>ov_msckf</code> with <code>colcon build</code> . Verify <code>rviz2</code> + ros2 daemon. | Finish the <b>8-Jun design deck</b> (proposal). Label all trajectory/drift charts <b>"illustrative / target — not measured."</b> Submit on time. |
| 11:00–13:00         | Run <b>OpenVINS on EuRoC V1_01_easy</b> out-of-box ( <code>ros2 launch ov_msckf subscribe.launch.py</code> ). Confirm pose stream on <code>/ov_msckf/odomimu</code> .       | Install <b>Gazebo Harmonic</b> + <code>ros-humble-ros-gzharmonic</code> bridge. Smoke-test an empty world + clock.                               |
| 14:00–16:00         | Record a rosbag of the EuRoC run ( <code>ros2 bag record /ov_msckf/odomimu /ov_msckf/trackcount</code> ). This bag is the <b>integrity safety-net source</b> .              | Stand up <b>PX4 SITL</b> via the <b>SathanBERNARD PX4-ROS2-Gazebo template</b> (or XTDrone); get a drone spawned + armed in a default world.     |
| 16:00–18:00         | Run <code>ov_eval</code> ATE on the EuRoC result; eyeball drift. First commit of <code>vio_core</code> launch configs.                                                      | Build a <b>straight tunnel/warehouse world</b> (textured walls — features matter). Confirm <code>/clock</code> , camera + IMU topics publish.    |
| <b>19:00 GATE 1</b> | <b>GO if:</b> OpenVINS runs on EuRoC and publishes <code>/ov_msckf/odomimu</code> . <b>If NO</b> → both swarm OpenVINS+EuRoC tomorrow AM; Gazebo slips to backup-only.      |                                                                                                                                                  |

### Day 2 — 9 Jun (LOCK THE FLOOR)

| Time        | Rayyan ( <code>vio_core</code> / <code>integrity</code> )                                                                                                                                       | Ashitha ( <code>sim</code> / <code>eval</code> / <code>deck</code> )                                                                                                                                                           |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 09:00–11:00 | Author <code>sensor_bridge</code> topic contract (remap Gazebo cam/IMU → OpenVINS expected topics + frame_ids).                                                                                 | Wire Gazebo tunnel → <code>sensor_bridge</code> ; fly a <b>scripted straight 200 m</b> pass; record ground-truth pose ( <code>/model/.../pose</code> or PX4 <code>/fmu</code> GT).                                             |
| 11:00–13:00 | Launch <b>OpenVINS on the Gazebo stream</b> ; tune IMU noise / cam rate until tracks lock.                                                                                                      | Record the <b>Gazebo VIO rosbag</b> (estimate + ground truth) — this is the deliverable artifact.                                                                                                                              |
| 14:00–16:00 | Start <code>integrity_monitor</code> (Python): subscribe <code>/ov_msckf/odomimu</code> (pose + covariance) and feature/track count; publish a <b>covariance-trace protection-level proxy</b> . | Run <code>evo_ape</code> / <code>evo_rpe --align_origin</code> + <code>ov_eval</code> <b>segment-RPE</b> on the Gazebo bag. Confirm <b>drift &lt; 1.5% over 200 m</b> ( $\leq 3.0$ m terminal). Generate estimate-vs-GT plots. |
| 16:00–18:00 | NEES check: integrity proxy vs sim ground truth (does the bound bound?). Commit <code>integrity_monitor</code> .                                                                                | Write the <b>estimate-vs-ground-truth performance report</b> ; clean + document the <b>ROS2 workspace</b> (README, build steps).                                                                                               |

| Time                     | Rayyan (vio_core / integrity)                                                                                                                                                                                                                                  | Ashitha (sim / eval / deck) |
|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|
| 19:00 GATE 2 — THE FLOOR | <b>LOCKED if ALL four DP7 deliverables exist:</b> (1) sim setup, (2) real-time VIO node, (3) perf report w/ drift < 1.5%, (4) documented workspace. <b>Tag the repo</b> <code>floor-locked</code> . From here, nothing on Days 3–4 may modify Floor artifacts. |                             |

**Floor fallback:** if Gazebo drift is shaky, the **EuRoC** ATE/RPE result satisfies deliverable (3) on real data, and stock OpenVINS-on-Gazebo + estimate-vs-GT still satisfies (1)(2)(4). The Floor does not depend on the integrity layer.

### Day 3 — 10 Jun (fenced UPSIDE: the live demo)

| Time         | Rayyan (integrity)                                                                                                                                                                                         | Ashitha (degradation / cockpit)                                                                                                                                               |
|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 09:00–12:00  | Harden the protection-level proxy: tune the <b>operational k=2.45</b> bound on covariance-trace; verify it tracks the EuRoC NEES.                                                                          | Build <code>degradation_manager</code> : subscribe integrity + track-count; implement <b>NOMINAL→DEGRADED→INERTIAL→RE-ACQUIRE</b> state machine on a continuous R(D) trigger. |
| 13:00–15:00  | <b>Kill-the-camera</b> mechanism: drop the camera topic (or <code>ros2 topic</code> mux off); confirm covariance blooms and proxy bound grows.                                                             | <code>health_cockpit</code> (Streamlit): trust HUD (green→red) + <b>breathing integrity ellipse</b> sized by the proxy bound, plotted over the moving truth dot.              |
| 15:00–17:00  | Tune the trigger so <b>trust</b> → <b>red in &lt; 0.5 s</b> of camera loss; verify bloomed ellipse <b>covers the true position 95%+</b> (the spine).                                                       | Wire it end-to-end; rehearse the <b>kill-the-camera beat sheet</b> (nominal → black → red → inertial → ellipse blooms ~1.0 m chasing & containing truth → recover).           |
| 17:00–18:30  | Run the integrity demo on the <b>EuRoC bag</b> too (sim-independent path) — the safety net.                                                                                                                | <b>Record the rosbag-replay</b> of a clean demo run (safety-ladder rung 2).                                                                                                   |
| 19:00 GATE 3 | <b>GO for live demo if:</b> trust flips red < 0.5 s on camera kill AND bloomed ellipse covers truth. <b>If NO</b> → demo from the recorded rosbag (rung 2); integrity math goes on slides as DERIVED only. |                                                                                                                                                                               |

### Day 4 — 11 Jun (de-risk, rehearse, backup video)

| Time         | Rayyan                                                                                                                                                                                                                                             | Ashitha                                                                                                     |
|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| 09:00–12:00  | <b>STRETCH only if GATE 3 green + time:</b> dump a rosbag and compute the full <b>observability index D</b> (whitened eigendecomposition) + <b>DAL-C k=6.74</b> dual bound <b>OFFLINE</b> → one backup plot. <i>Do not</i> fork OpenVINS C++ live. | Polish cockpit visuals; finalize the <b>three-rung safety ladder</b> (live → rosbag-replay → backup video). |
| 13:00–15:00  | Freeze all code. Re-verify EuRoC + Gazebo demos still run from clean checkout of the <code>floor-locked</code> tag + upside branch.                                                                                                                | <b>Record the backup demo video</b> (rung 3) — narrate the spine line over it.                              |
| 15:00–17:00  | Joint: full <b>dry-run on real hardware</b> the team brings on the 12th. Time the 3-min beat sheet.                                                                                                                                                | Bake <b>offline-derived D / DAL-C plot</b> into the backup slide; label math "derived + computed offline."  |
| 17:00–18:00  | Buffer / bugfix. Tag <code>demo-final</code> .                                                                                                                                                                                                     | Buffer / deck polish. Print the beat-sheet cue card.                                                        |
| 19:00 GATE 4 | <b>Demo-ready if:</b> at least one rung of the ladder runs clean end-to-end on the demo machine. (Floor is already locked, so this gate only governs the <i>live</i> upside, never the deliverables.)                                              |                                                                                                             |

### Notes & guardrails

- Why the fence matters:** the Floor (Day 2) is 100% of the DP7 deliverables and ~70% of the rubric (Problem Understanding, Planning, Presentation). The Upside (Days 3–4) is the Innovativeness multiplier. Never trade a locked deliverable for an unlocked stretch.
- The "say it / run it" split:** *say* the full RAIM-analogue math (fault-hypothesis bound, observability index D, dual k=2.45/6.74 bound) on slides; *run* the Python covariance-trace proxy live. The C++ `UpdaterMSCKF::update` fork is **slide-only, offline-derived** — explicitly never on the Day-3/4 critical path.
- 2nd-year + AI realism:** every block is "integrate a named, documented tool" (OpenVINS, evo, ov\_eval, Streamlit) — not invent an algorithm. AI assist is for glue code, launch files, and Streamlit, not the MSCKF core (use OpenVINS as-is).
- Single biggest schedule killer:** the OS migration (Day 0) and OpenVINS-on-Gazebo topic plumbing (Day 2 AM). Both have explicit gates that fall back to **EuRoC-only**, which alone satisfies all four deliverables.
- Commit discipline:** tag `floor-locked` (end Day 2) and `demo-final` (end Day 4). The 12 Jun demo machine checks out `demo-final` ; if anything breaks, `git checkout floor-locked` still gives a clean, scoring submission.

## 12. Evaluation & Performance-Report Spec

This section defines exactly how A.E.T.H.E.R measures itself against the DP7 accuracy target and how the deliverable "estimate-vs-ground-truth performance report" is produced. Every number is computed from logged rosbags + ground truth with `ov_eval` and `evo` — no eyeballing. **The 8-**

Jun design deck shows no measured numbers; every trajectory/drift/ellipse chart on that deck is labelled ILLUSTRATIVE / TARGET . Measured numbers land only after the 9–11 Jun build and replace these placeholders in the final report.

What DP7 actually asks for, and the metric trap

DP7 says "drift < 1.5% over 200 m." That is the **KITTI segment-relative metric**, not a single end-point error. The trap is reporting end-to-end ATE and calling it "drift %" — they are different quantities and a strong VIO can pass one and fail the other. We report both, but the headline is the KITTI segment %.

- **Terminal-error reading of the spec:**  $1.5\% \times 200\text{ m} = 3.0\text{ m}$  allowable terminal error. We use  $\leq 3.0\text{ m}$  terminal ATE as a secondary sanity gate.
- **Primary metric (what we headline):** KITTI segment-RPE %drift, averaged over segment lengths **{10, 20, 40, 80} m** (the standard KITTI sub-segment set that fits inside a 200 m run; KITTI's full {100...800} set does not fit and is not used).

KITTI segment %drift — how it is computed correctly

The KITTI segment metric slides a window of length  $L$  along the ground-truth path, aligns the estimate to ground truth **at the start of each segment only** (not globally), measures the translational error accumulated over that segment, and normalizes by  $L$ . Averaging over many start points and over  $L \in \{10,20,40,80\}$  m gives a single %drift that is insensitive to where the run happens to end.

We do **not** hand-roll this. `ov_eval` implements it directly against the OpenVINS convention:

```
# Per-run KITTI segment error (the DP7 headline number)
rosrun ov_eval error_singlerun none groundtruth.txt estimate.txt
# -> prints RMSE ATE, and segment errors for {10,20,40,80} m as (% , deg/m)

# Cross-check the same quantity with evo's relative-pose error:
evo_rpe tum groundtruth.txt estimate.txt \
  --delta 40 --delta_unit m --all_pairs \
  --align_origin -v --plot --save_results rpe_40m.zip
```

`--all_pairs` makes `evo_rpe` evaluate **every** pair separated by  $L$ , matching the KITTI all-start-points convention; the per- $L$  mean translation error divided by  $L$  is the %drift for that segment length. We run it for each  $L$  and report the mean.

evo\_ape vs evo\_rpe — what each one is for

| Tool                                                 | Quantity                          | Alignment                                       | What it answers                                     | Role in report                              |
|------------------------------------------------------|-----------------------------------|-------------------------------------------------|-----------------------------------------------------|---------------------------------------------|
| <code>evo_ape</code>                                 | Absolute Pose Error (global)      | <code>--align_origin</code> (origin + yaw only) | "How far is the whole trajectory from truth?"       | Secondary: terminal/RMS ATE, the 3.0 m gate |
| <code>evo_rpe</code>                                 | Relative Pose Error over $\Delta$ | per-pair, local                                 | "How much does it drift <i>per unit distance</i> ?" | Tightly tracks the KITTI %drift headline    |
| <code>ov_eval</code><br><code>error_singlerun</code> | KITTI segment % + NEES + ATE      | per-segment-start                               | The official DP7 drift number + filter consistency  | <b>Primary</b>                              |

The `--align_origin` honesty rule

There are three alignment modes, and the choice changes the number dramatically. We are explicit about which we use and why:

- `--align` (full SE(3) Umeyama, optionally `-s` for scale): rigidly fits the *entire* estimated trajectory onto truth. **Forbidden for the headline.** It hides drift by rotating the whole path to minimize global error, and `-s` (scale alignment) is outright dishonest for a metric-scale stereo system — it would mask scale drift, the exact failure VIO is supposed to avoid.
- `--align_origin` : aligns **only the first pose** (position + yaw). This is the honest VIO convention — the drone starts at a known origin/heading and everything after is pure dead-reckoning error. **This is what we use for every ATE/RPE plot.** Stereo gives metric scale, so we never scale-align.
- KITTI segment metric: aligns per-segment-start internally, so global alignment never enters the headline at all.

Honesty statement printed in the report: "All ATE/RPE figures use `--align_origin` (first-pose alignment, no scale). No global Umeyama or scale alignment is used anywhere. Stereo VIO is metric-scale; scale drift is reported, not removed."

Held-out split — tune vs report

To avoid the "tuned on the test set" critique a Honeywell SME will absolutely make, we separate tuning from reporting at the **Gazebo seed** level:

| Role                 | Environment / seeds            | Used for                                                                                                                        |
|----------------------|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| <b>Tuning set</b>    | Warehouse, seeds <b>{1, 2}</b> | OpenVINS noise params (gyro/accel $\sigma$ , feature count, MSCKF window), degradation thresholds, integrity-bound constant $k$ |
| <b>Held-out test</b> | Tunnel, seed <b>{3}</b>        | Every reported number. Frozen config — no parameter touched after first run on seed 3                                           |



| Role              | Environment / seeds                                         | Used for                                                                   |
|-------------------|-------------------------------------------------------------|----------------------------------------------------------------------------|
| External validity | EuRoC MAV ( <a href="#">MH_03</a> , <a href="#">V1_02</a> ) | Confirms the same frozen config works on real-world data, not just our sim |

The config hash (git SHA of the [ov\\_msckf](#) YAML) is printed on every plot so a judge can confirm tuning and reporting used the **same** frozen params.

Filter consistency — NEES (proving the estimator isn't lying)

A VIO can be accurate *and* overconfident; for an integrity claim that is fatal. NEES (Normalized Estimation Error Squared) checks whether the reported covariance actually matches the true error:

NEES\_k = e\_k^T · P\_k^{-1} · e\_k ,    e\_k = x\_k - x\_k(truth)

For a consistent filter the time-averaged NEES sits near the state dimension *d*, inside the two-sided 95% chi-square bounds. [ov\\_eval](#) [error\\_singlerun](#) emits the pose NEES directly against sim ground truth.

- **Pass band (d = 6, pose):** ANEES ∈ [4.6, 7.8] approx ( $\chi^2_6$  95% two-sided / DOF, batch-averaged).
- NEES **above** the band ⇒ overconfident (covariance too small ⇒ protection bound too tight ⇒ unsafe). NEES **below** ⇒ conservative (safe but loose). For an integrity system, *conservative is acceptable, overconfident is a defect*.

Integrity-bound (IB) coverage — the novelty's pass/fail

The integrity layer publishes a live **protection-level proxy** (the breathing bound) derived from the pose covariance trace on [/ov\\_msckf/odomimu](#) plus tracked-feature count. The claim — "*the bound covers true position 95%+ of the time*" — is tested as a coverage rate against sim ground truth, with a **dual bound**:

| Bound                 | k    | Source                                                                                                                     | Coverage target         | Use                               |
|-----------------------|------|----------------------------------------------------------------------------------------------------------------------------|-------------------------|-----------------------------------|
| Operational           | 2.45 | 95% 2-D (Rayleigh-ish) operating point                                                                                     | ≥ 95% of samples inside | The breathing ellipse in the demo |
| DAL-C (DO-178C-as-if) | 6.74 | <a href="#">k_ffd</a> from $\lambda_{md} = 45.0$ , <code>scipy.stats</code> — fault-free detection at DAL-C integrity risk | ≥ 99.999% inside        | The certification-framed slide    |

scipy verification snippet (run once, numbers pinned in the report):

```
from scipy.stats import norm, ncx2
# DAL-C fault-free-detection multiplier
P_HMI = 1e-7 # integrity risk budget, DAL-C-as-if
k_ffd = norm.ppf(1 - P_HMI/2) # -> 6.74 (this is the 6.74)
# missed-detection non-centrality for the slope check
lambda_md = 45.0 # pinned design point
# Reported: operational k=2.45, DAL-C k=6.74; IB ~1.31 m covers true ~0.3 m error
```

**Coverage metric:** fraction of time-steps where  $\|\hat{p} - p_{truth}\| \leq IB(k)$  . Reported per bound:

| Quantity                    | Operational k=2.45 | DAL-C k=6.74 |
|-----------------------------|--------------------|--------------|
| Empirical coverage (target) | ≥ 95%              | ≥ 99.999%    |
| Mean IB radius              | ≈ 1.31 m           | (larger)     |
| Mean true error contained   | ≈ 0.3 m            | ≈ 0.3 m      |

The headline integrity sentence — "*IB ≈ 1.31 m covers the true ≈ 0.3 m error*" — means the bound is **valid (covers truth) but not loose** (only ~4× the actual error), which is exactly the RAIM property we claim. NEES-pass + coverage-pass together are the proof that the bound bounds.

Exact list of report figures

The performance report (PDF + in the documented ROS2 workspace [docs/](#) ) contains exactly these, each captioned with run = {*env, seed, config-SHA, date*} :

1. **Trajectory overlay (top-down XY):** estimate vs ground truth, [--align\\_origin](#) , tunnel seed 3.
2. **3-D trajectory overlay:** same run, perspective view (tunnel has vertical structure).
3. **ATE-over-time:** [evo\\_ape](#) error curve, with the 3.0 m terminal gate drawn as a horizontal line.
4. **KITTI segment %drift bar chart:** one bar per  $L \in \{10,20,40,80\}$  m + the 1.5% target line. **The DP7 money figure.**
5. **RPE histogram** ( [evo\\_rpe --delta 40 m --all\\_pairs](#) ): distribution of per-segment drift.
6. **NEES-over-time** with the  $\chi^2_6$  95% band shaded — the consistency proof.
7. **IB-coverage timeline:**  $\|\hat{p} - p_{truth}\|$  vs operational and DAL-C bounds; shaded "inside/outside."

- 8. **The kill-the-camera panel:** trust state + breathing-ellipse radius vs time across the nominal→black→inertial→re-acquire beat sheet; annotate <0.5 s detection and the bloom-and-contain.
- 9. **Drift-vs-outage-duration curve:** modeled INS drift for 2/4/8 s vision outages on HG4930-class IMU (≈0.4 m @ 8 s), cross-checked to physics ±20% — the Honeywell hook, labelled DERIVED.
- 10. **EuRoC external-validity row:** ATE + %drift on `MH_03` / `V1_02` to show it isn't sim-only.

Headline metrics table (the single slide that wins the rubric)

This is the table the report leads with and the deck closes on. On 8 Jun every cell reads `target` and is labelled illustrative; on 12 Jun it is replaced with measured values from the frozen-config tunnel seed-3 run.

| Metric                               | DP7 / design target    | Tool                                        | 8-Jun status | Final (12-Jun)  |
|--------------------------------------|------------------------|---------------------------------------------|--------------|-----------------|
| KITTI segment %drift {10,20,40,80} m | < 1.5 %                | <code>ov_eval</code> / <code>evo_rpe</code> | target       | <i>measured</i> |
| Terminal ATE over 200 m              | ≤ 3.0 m                | <code>evo_ape --align_origin</code>         | target       | <i>measured</i> |
| RMS ATE (full run)                   | ≤ ~1.5 m               | <code>evo_ape</code>                        | target       | <i>measured</i> |
| Pose ANEES (consistency)             | ∈ [4.6, 7.8] (d=6)     | <code>ov_eval</code>                        | target       | <i>measured</i> |
| IB coverage @ k=2.45                 | ≥ 95 %                 | <code>integrity_monitor</code>              | target       | <i>measured</i> |
| IB coverage @ k=6.74 (DAL-C)         | ≥ 99.999 %             | <code>integrity_monitor</code>              | target       | <i>measured</i> |
| Mean IB radius vs true error         | ≈ 1.31 m vs ≈ 0.3 m    | <code>integrity_monitor</code>              | target       | <i>measured</i> |
| Camera-fault detection latency       | < 0.5 s                | <code>degradation_manager</code>            | target       | <i>measured</i> |
| INS-only drift @ 8 s outage          | ≈ 0.4 m (HG4930-class) | derived model                               | DERIVED      | DERIVED         |

**Reading guide for judges (printed under the table):** the first three rows answer DP7's literal accuracy ask; rows 4–7 are the integrity novelty (the estimator is not only accurate but *honest about its own uncertainty*); rows 8–9 are the operational/Honeywell story. A green first block alone satisfies all four DP7 deliverables; the rest is why A.E.T.H.E.R is integrity-*aware*, not just another VIO.

13. ROS2 Workspace Layout, Message Contract & Glossary

This section pins the physical layout of the A.E.T.H.E.R ROS2 workspace, the custom message types that carry the integrity layer's outputs, the launch-file topology that makes "any subset runs" true, and a plain-English glossary so a navigation SME can read our code without a decoder ring. Everything below targets **ROS2 Humble** on **Ubuntu 22.04**.

Reality check: the **guaranteed-floor** build is stock OpenVINS + the Python `integrity_monitor` consuming what `ov_msckf` already publishes ( `/ov_msckf/odomimu` , feature counts). The C++ message internals ( `FeatureInternals.msg` , the forked publisher) are the **stretch** path — defined here so the contract is documented, computed **offline on a dumped rosbag** if Day-4 time exists, not bet on for the live demo.

## Workspace tree

```
~/sentinel_ws/
├─ src/
│   ├── sentinel_msgs/           # custom .msg/.srv - built FIRST (everything deps on it)
│   │   ├── msg/
│   │   │   ├── VioInternals.msg  # vio_core -> integrity_monitor (stretch: forked OpenVINS)
│   │   │   ├── IntegrityState.msg # integrity_monitor -> degradation_manager + cockpit
│   │   │   ├── ProtectionLevel.msg # the breathing bound (RAIM analogue on segment-RPE)
│   │   │   └─ NavMode.msg        # NOMINAL / DEGRADED / INERTIAL / RE_ACQUIRE enum
│   │   ├── srv/
│   │   │   └─ KillCamera.srv     # demo trigger: blank the image stream on command
│   │   ├── CMakeLists.txt        # rosidl_generate_interfaces(...)
│   │   └─ package.xml            # buildtool rosidl_default_generators; exec rosidl_default_runtime
│   └─ sentinel_bringup/         # launch + params + rviz config (no nodes - pure orchestration)
│       ├── launch/
│       │   ├── sim_world.launch.py # Gazebo Harmonic tunnel + PX4 SITL + ros_gz bridge
│       │   ├── vio.launch.py       # ov_msckf + sensor_bridge (the guaranteed floor)
│       │   ├── integrity.launch.py  # integrity_monitor + degradation_manager + cockpit
│       │   └─ sentinel.launch.py    # top-level: includes all three above (full system)
│       ├── config/
│       │   ├── estimator_config.yaml # OpenVINS params (cam/imu intrinsics, MSCKF window)
│       │   ├── integrity.yaml        # k_op=2.45, k_ffd=6.74, lambda_md=45.0, NEES window N
│       │   └─ sentinel.rviz         # ellipse Marker + trajectory + ground-truth display
│       └─ package.xml              # ament_cmake, exec_depend on every runtime pkg
│   └─ sensor_bridge/              # ament_python: remaps ros_gz / EuRoC topics -> OpenVINS in
│       ├── sensor_bridge/bridge_node.py
│       └─ package.xml setup.py setup.cfg
│   └─ integrity_monitor/          # ament_python: THE NOVEL LAYER (Python proxy = live demo)
│       ├── integrity_monitor/
│       │   ├── monitor_node.py     # subscribes /ov_msckf/odomimu (+feature cnt) -> IntegrityState
│       │   ├── protection_level.py # cov-trace / PL proxy -> ProtectionLevel (the "ellipse")
│       │   ├── nees.py             # NEES vs sim ground truth (proves the bound bounds)
│       │   └─ raim.py              # OFFLINE: fault-hypothesis k-factor bound (stretch slide)
│       └─ package.xml setup.py setup.cfg
│   └─ degradation_manager/        # ament_python: continuous R(D) state machine
│       ├── degradation_manager/manager_node.py # IntegrityState -> NavMode transitions
│       └─ package.xml setup.py setup.cfg
│   └─ health_cockpit/             # ament_python: Streamlit trust HUD + RViz Marker publisher
│       ├── health_cockpit/cockpit_node.py      # NavMode+PL -> breathing ellipse Marker + HUD
│       └─ package.xml setup.py setup.cfg
├─ bags/                          # dumped rosbag2 for offline RAIM + backup-rung replay
└─ results/                       # evo / ov_eval plots (ATE, segment-RPE, NEES) for the report
```

Build order is forced by dependency: `sentinel_msgs` (CMake/rosidl) → the Python nodes → `sentinel_bringup`. Mixed build types are fine in one workspace; `colcon` resolves order from `package.xml` `<depend>` tags.

```
cd ~/sentinel_ws
colcon build --symlink-install --packages-select sentinel_msgs # interfaces first
colcon build --symlink-install                               # the rest
source install/setup.bash
```

## Custom message definitions

These live in `sentinel_msgs/msg/`. All units SI; frames follow REP-103 (x-forward, y-left, z-up) and REP-105 (`odom`, `base_link`). The **field names are the Honeywell-facing contract** — `protection_level`, `observability_index`, `nav_mode` map one-to-one to the RAIM / INS-aiding vocabulary the judges already speak.

`VioInternals.msg` — emitted by `vio_core` (stretch: forked `ov_msckf` ; floor: a thin shim reposting `/ov_msckf/odomimu` covariance):

```
std_msgs/Header header          # stamp = IMU time, frame_id = "odom"
geometry_msgs/PoseWithCovariance pose  # 6x6 cov = the state uncertainty we bound
float64[3] velocity             # body-frame linear velocity (INS spine state)
uint32 num_slam_features        # features in the long-term SLAM map
uint32 num_msckf_features       # features used in this MSCKF update
float64 cond_number             # condition # of the update info matrix (observability proxy)
float64 disparity_px            # median feature disparity – vision-aiding strength
```

**ProtectionLevel.msg** — the breathing bound; this is the GNSS-RAIM protection-level analogue applied to the **segment-RPE** quantity DP7 grades:

```
std_msgs/Header header
float64 pl_operational          # k_op=2.45 bound (~95% / operational)
float64 pl_dal_c               # k_ffd=6.74 bound (DAL-C / D0-178C-as-if)
float64 covariance_trace       # sqrt(trace) of position cov – the live "ellipse" radius
float64[3] semi_axes           # 3-sigma position-error ellipsoid semi-axes (RViz Marker)
float64 horizontal_pl          # 2D PL for the cockpit's flat map view
```

**IntegrityState.msg** — the monitor's verdict, the bus the manager and cockpit both read:

```
std_msgs/Header header
sentinel_msgs/ProtectionLevel protection
float64 observability_index    # D in [0,1]: 1=fully observable, 0=unobservable direction
float64 nees                   # normalized estimation error squared vs sim ground truth
float64 nis                    # normalized innovation squared (per-update consistency)
bool bound_valid              # NEES inside chi-square gate -> bound is trustworthy
float64 time_since_aiding_s    # seconds since last accepted vision update (outage clock)
sentinel_msgs/NavMode mode
```

**NavMode.msg** — the degradation enum, stringly-safe via named constants:

```
uint8 NOMINAL    = 0    # vision + IMU, bound tight
uint8 DEGRADED   = 1    # vision weak (low features / high NIS), bound widening
uint8 INERTIAL    = 2    # vision dead, INS-only dead-reckon, ellipse blooming
uint8 RE_ACQUIRE = 3    # vision returning, bound contracting back to NOMINAL
uint8 mode
```

**KillCamera.srv** — the demo's "kill-the-camera" beat, callable from CLI so the operator triggers it on stage:

```
bool enable      # true = blank the image stream
---
bool acknowledged
```

```
ros2 service call /kill_camera sentinel_msgs/srv/KillCamera "{enable: true}"
```

## Launch-file structure (independence by construction)

The launch graph is layered so **any subset runs** — the bible's non-negotiable. Each layer degrades gracefully if the one below is absent (a missing topic just freezes a HUD field; nothing crashes).

| Launch file                      | Brings up                                                                                       | Runs without                                              |
|----------------------------------|-------------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| <code>sim_world.launch.py</code> | Gazebo Harmonic tunnel, PX4 SITL, <code>ros_gz_bridge</code>                                    | — (or skip entirely and feed EuRoC)                       |
| <code>vio.launch.py</code>       | <code>sensor_bridge</code> , <code>ov_msckf</code>                                              | the integrity layer (this alone = all 4 DP7 deliverables) |
| <code>integrity.launch.py</code> | <code>integrity_monitor</code> , <code>degradation_manager</code> , <code>health_cockpit</code> | live sim — replays a rosbag instead (safety net)          |
| <code>sentinel.launch.py</code>  | includes all three above                                                                        | — (full stage demo)                                       |

```
# sentinel.launch.py – top-level composition
from launch import LaunchDescription
from launch.actions import IncludeLaunchDescription, DeclareLaunchArgument
from launch.substitutions import LaunchConfiguration, PathJoinSubstitution
from launch.conditions import IfCondition
from launch_ros.substitutions import FindPackageShare

def generate_launch_description():
    pkg = FindPackageShare('sentinel_bringup')
    use_sim = LaunchConfiguration('use_sim') # false => EuRoC/rosbag (the safety net)
    return LaunchDescription([
        DeclareLaunchArgument('use_sim', default_value='true'),
        IncludeLaunchDescription(
            PathJoinSubstitution([pkg, 'launch', 'sim_world.launch.py']),
            condition=IfCondition(use_sim)),
        IncludeLaunchDescription(PathJoinSubstitution([pkg, 'launch', 'vio.launch.py'])),
        IncludeLaunchDescription(PathJoinSubstitution([pkg, 'launch', 'integrity.launch.py'])),
    ])

```

```
# Three-rung safety ladder, one flag each:
ros2 launch sentinel_bringup sentinel.launch.py use_sim:=true # rung 1: live sim
ros2 launch sentinel_bringup sentinel.launch.py use_sim:=false # rung 2: rosbag/EuRoC replay
ros2 bag play bags/euroc_kill_demo --clock # (paired with rung 2)
# rung 3 = a recorded screen-capture .mp4 – no ROS2 needed

```

## The documented-source story (the 4th DP7 deliverable)

DP7 explicitly grades a **"documented ROS2 workspace."** We satisfy it by construction, not by a doc sprint on 11 Jun:

- **Per-package** `README.md` — each of the 6 packages states its single responsibility, its subscribed/published topics with message types, and its run command. The interface table is the README.
- **Provenance is explicit.** Stock upstream ( `ov_msckf` , `ros_gz_bridge` , PX4 SITL) is pulled by `vcs import` from a checked-in `sentinel.repos` file; our novelty lives only in `integrity_monitor` , `degradation_manager` , `health_cockpit` . A reviewer sees exactly which lines are ours.
- **Params not magic numbers.** Every tunable ( `k_op` , `k_ffd` , `lambda_md` , NEES window, MSCKF clone count) is in a YAML under `config/` , loaded via `declare_parameter` , with an inline comment citing its source (scipy verification or the OpenVINS default).
- **Topic graph in the report.** `rqt_graph` export + `ros2 topic list -t` dump go straight into the performance report, next to the evo/ov\_eval plots, so the node→message→node flow is auditable.
- **Reproducibility.** One `colcon build` after `vcs import` ; one `ros2 launch` per safety rung; the EuRoC path needs no GPU, so a judge can re-run the integrity demo on a laptop.

## Glossary (plain English)

| Term                                                   | Plain-English meaning                                                                                                                                                                                                                                                 |
|--------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>VIO</b> (Visual-Inertial Odometry)                  | Estimating where you are by fusing a camera (what moved in the image) with an IMU (accelerometer + gyro). No GPS. Our whole localizer.                                                                                                                                |
| <b>MSCKF</b> (Multi-State Constraint Kalman Filter)    | The specific VIO algorithm in OpenVINS. Instead of carrying every 3D landmark in the filter state, it keeps a sliding window of recent camera poses and uses each feature as a geometric constraint across them — accurate and cheap.                                 |
| <b>IMU</b>                                             | Inertial Measurement Unit: 3-axis accelerometer + 3-axis gyroscope. The "spine" — always available, drifts over time, bounded by the vision aid.                                                                                                                      |
| <b>ATE</b> (Absolute Trajectory Error)                 | After aligning your estimated path to ground truth, the overall position error of the whole trajectory. One number for "how good is the map."                                                                                                                         |
| <b>RPE / segment-RPE</b> (Relative Pose Error)         | Error accumulated over a fixed segment length (e.g. per 100 m), not over the whole path. This is the <b>drift</b> metric DP7 grades — it's why $<1.5\%$ over 200 m $\Rightarrow \leq 3.0$ m.                                                                          |
| <b>drift %</b>                                         | RPE expressed as a fraction of distance travelled. 1.5% means 1.5 m of error for every 100 m flown. The headline DP7 number.                                                                                                                                          |
| <b>NEES</b> (Normalized Estimation Error Squared)      | A consistency check: actual error <sup>2</sup> scaled by the filter's claimed covariance. If the filter says "I'm confident" it had better <i>be</i> right — NEES near the expected chi-square value proves the covariance (and our bound) is honest, not optimistic. |
| <b>NIS</b> (Normalized Innovation Squared)             | Same idea but per measurement update: how surprising was each new observation given the predicted uncertainty. A spike = a bad/faulted feature — our per-update fault detector.                                                                                       |
| <b>RAIM</b> (Receiver Autonomous Integrity Monitoring) | A GPS concept: the receiver checks its <i>own</i> solution for faults and outputs a guaranteed error bound. <b>Our core analogy</b> — we do RAIM, but for the <i>vision</i> aid instead of satellites.                                                                |

| Term                                  | Plain-English meaning                                                                                                                                                                                                                                                            |
|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Protection Level</b>               | RAIM's output: a number guaranteeing "your true error is inside this bound with X% probability." Our breathing ellipse is a protection level on the segment-RPE quantity.                                                                                                        |
| <b>observability (index D)</b>        | Whether the sensors can actually <i>see</i> a given state direction. A drone flying straight down a featureless tunnel can't observe scale/lateral drift well. D in [0,1] tells us <i>which</i> directions vision is failing to constrain — so we widen the bound exactly there. |
| <b>graceful degradation</b>           | The system gets <i>worse safely</i> instead of failing hard. Vision dies → drop to INERTIAL, bloom the bound, keep flying — never a silent wrong answer. Built into the launch graph: any subset of nodes still runs.                                                            |
| <b>FEJ</b> (First-Estimates Jacobian) | A trick to keep an EKF/MSCKF consistent: evaluate the linearization at the <i>first</i> estimate of each state so the filter doesn't illegally "gain information" it never observed. Keeps NEES honest.                                                                          |
| <b>ZUPT</b> (Zero-velocity Update)    | When the platform is known stationary, feed the filter a "velocity = 0" measurement to crush IMU drift. Cheap integrity boost while hovering/landed.                                                                                                                             |
| <b>Dead-reckoning</b>                 | Propagating position from IMU alone (integrate acceleration twice) with no external aid. What INERTIAL mode does when the camera is dead.                                                                                                                                        |

### References

**Algorithms & filters** - Geneva, Eckenhoff, Lee, Huang, *OpenVINS: A Research Platform for Visual-Inertial Estimation*, ICRA 2020. ( `ov_msckf` , `ov_eval` — our VIO core + evaluator.) - Mourikis & Roumeliotis, *A Multi-State Constraint Kalman Filter for Vision-Aided Inertial Navigation*, ICRA 2007. (The MSCKF.) - Qin, Li, Shen, *VINS-Mono / VINS-Fusion*, IEEE T-RO 2018. (Backup VIO.) - Campos et al., *ORB-SLAM3*, IEEE T-RO 2021. (Benchmark, cited for comparison.) - Forster, Carlone, Dellaert, Scaramuzza, *On-Manifold Preintegration for Real-Time VIO*, IEEE T-RO 2017. (IMU preintegration — GTSAM.)

**Observability, consistency & integrity (the novelty's literature)** - Hesch, Kottas, Bowman, Roumeliotis, *Observability-Constrained VINS (OC-EKF)*, IJRR 2014. (Grounds FEJ / observability index D.) - Zhang & Singh, *On Degeneracy of Optimization-based State Estimation Problems*, ICRA 2016. (Degeneracy / observability index basis.) - Gupta & Gao, *Data-Driven Protection Levels for the Integrity of Visual Navigation*, ION NAVIGATION. (The literature gap we fill: integrity bounds for vision.) - RTCA DO-178C, *Software Considerations in Airborne Systems* — DAL-C framing (DO-178C-as-if).

**Datasets, sim & tooling** - Burri et al., *The EuRoC MAV Dataset*, IJRR 2016. (Out-of-box VIO validation + GPU-free integrity demo.) - Grupp, *evo: Python package for trajectory evaluation* ( `evo_ape` , `evo_rpe` , `--align_origin` ). - ROS2 Humble; Gazebo Harmonic + `ros-humble-ros-gzharmonic` ; PX4 SITL; SathanBERNARD *PX4-ROS2-Gazebo* template; XTDrone. - `scipy.stats` — chi-square / k-factor verification ( `lambda_md=45.0` , `k_ffd=6.74` , `k_op=2.45` ); `robot_localization` (stretch UWB); XFeat / SuperPoint (stretch front-end); Streamlit (trust HUD).

**Honeywell / hardware** - Honeywell **HG4930 CA51** MEMS IMU — datasheet (the tactical-grade spine; 8 s outage ⇒ ~0.4 m drift). - Honeywell **HANA** (Honeywell Alternative Navigation Architecture, announced Oct 2025; first release = vision-aided nav) — the industrial analogue we frame against.

## 14. Risk Register & SME Q&A Answer Bank

This section is the team's pre-mortem and its defense. Part 1 is the live risk register we track daily through the 8–11 Jun build window. Part 2 is the answer bank for the kind of pointed, kill-shot question a Honeywell navigation SME asks when they want to find out whether you actually understand what you built or just wired tools together.

### Part 1 — Expanded Risk Register

Severity = impact if it fires. Likelihood is post-mitigation. "Owner" reflects the two-person split (R = Rayyan, A = Ashitha). Every row has a *concrete* fallback that keeps at least the DP7 guaranteed floor intact.

| #  | Risk                                                                                                                           | Sev  | Likelihood | Trigger / how we detect it                                             | Mitigation                                                                                                                                                                                           | Fallback if it fires                                                                                                                                                                   |
|----|--------------------------------------------------------------------------------------------------------------------------------|------|------------|------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| R1 | <b>Windows→Ubuntu environment loss</b> — ROS2 Humble + Gazebo Harmonic + PX4 need Ubuntu 22.04; WSL2/VM GPU+timing is fragile. | High | Med        | Day-0 (8 Jun): if native dual-boot isn't booting ROS2 + Gazebo by EOD. | <b>Day-0 must-do:</b> native dual-boot Ubuntu 22.04, then <code>apt install ros-humble-desktop ros-humble-ros-gzharmonic</code> . Verify with <code>ros2 topic list</code> against a running Gazebo. | Cloud GPU box (Lambda/Paperspace, Ubuntu 22.04 image) provisioned same day; or fall to WSL2 for the <i>integrity Python node only</i> (no Gazebo render needed — it runs on a rosbag). |

| #  | Risk                                                                                                                                                            | Sev  | Likelihood     | Trigger / how we detect it                                                                                              | Mitigation                                                                                                                                                                                                                                                                                                                           | Fallback if it fires                                                                                                                                                                                                             |
|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|------|----------------|-------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| R2 | <b>OpenVINS won't converge on our Gazebo tunnel</b> — scale drift, filter divergence, or features starve in low-texture corridor.                               | High | Med            | <code>ov_eval</code> segment-RPE > 1.5% on the Gazebo run, or <code>/ov_msckf/odomimu</code> covariance trace diverges. | Tune <code>ov_msckf</code> config (feature count <code>num_pts</code> , MSCKF clone window, IMU noise from sim). Add wall texture/posters to the Gazebo tunnel to feed the tracker. Validate the <i>exact same config</i> converges on EuRoC first.                                                                                  | <b>EuRoC is the safety net:</b> the entire integrity demo runs on EuRoC (real, hard data) independent of Gazebo. Gazebo deliverable is then satisfied with stock VIO + estimate-vs-GT plots only, not the live kill-camera beat. |
| R3 | <b>C++ fork of OpenVINS ( <code>UpdaterMSCKF::update</code> ) eats the build</b> — publishing per-feature internals breaks the filter or won't compile in time. | Med  | Low (descoped) | Any time the fork branch isn't merged-and-running by Day-4 morning.                                                     | <b>It is explicitly out of the critical path.</b> The live demo uses only what stock <code>ov_msckf</code> already publishes: pose+covariance on <code>/ov_msckf/odomimu</code> and tracked-feature count.                                                                                                                           | Present the full per-feature NIS / RAIM-slope math as <b>DERIVED, computed OFFLINE</b> on a dumped rosbag — one backup slide + one offline plot. "Say the full math; run the Python proxy."                                      |
| R4 | <b>Calibration / time-sync error</b> — cam-IMU extrinsics or timestamp offset wrong → VIO diverges in a way that <i>looks</i> like our integrity layer failing. | Med  | Med            | Sudden RPE blow-up uncorrelated with motion; NEES persistently > $\chi^2$ bound everywhere (not just at faults).        | In sim, extrinsics and <code>t_off</code> are <i>known exactly</i> from the SDF/PX4 model — we read them, we don't estimate them. Keep <code>calib_cam_extrinsics</code> / <code>calib_cam_timeoffset</code> estimation <b>off</b> for the sim run. On EuRoC, use the published Kalibr calibration shipped with the dataset.         | If sim sync is suspect, fall back to EuRoC entirely (R2 net). Time-sync sanity check: cross-correlate IMU                                                                                                                        |
| R5 | <b>Live demo unreliable on stage</b> — Gazebo render hitch, node crash, or laptop thermal throttle during the 3-hour final.                                     | High | Med            | Dry-run on 11 Jun stutters, or any node drops a heartbeat on <code>/diagnostics</code> .                                | <b>Three-rung safety ladder</b> , rehearsed in order: (1) live, (2) <code>ros2 bag play</code> of a pre-recorded golden run, (3) backup screen-recording video. The kill-camera beat is identical on rungs 1 and 2.                                                                                                                  | Drop to rung 2 (rosbag replay) on first stutter — the integrity_monitor consumes the bag identically, so the breathing ellipse + trust-red beat plays the same. Rung 3 video is the floor.                                       |
| R6 | <b>2-person bandwidth</b> — both 2nd-years, AI-assisted; one person blocked stalls the whole pipeline.                                                          | Med  | Med            | Any owned node slips a day vs the 8–11 build schedule.                                                                  | <b>Module independence by construction:</b> <code>sensor_bridge</code> / <code>vio_core</code> / <code>integrity_monitor</code> / <code>degradation_manager</code> / <code>health_cockpit</code> each run standalone; any subset = graceful degradation. R owns sim+VIO bring-up; A owns integrity Python + HUD. Daily merge at EOD. | If one half slips, the other half still demos: stock VIO + GT plots (R's half) <i>or</i> integrity-on-EuRoC (A's half) each independently satisfy part of DP7.                                                                   |
| R7 | <b>Drift target missed (&gt;1.5% / &gt;3.0 m terminal)</b> on the headline run.                                                                                 | High | Low            | <code>ov_eval</code> / <code>evo_rpe</code> final numbers above threshold.                                              | Validate the config on EuRoC (where OpenVINS is <i>known</i> to hit <1% on MH sequences) before trusting Gazebo numbers. Use <code>evo_ape --align_origin</code> and segment-RPE per DP7's KITTI metric.                                                                                                                             | Report the <i>best converged</i> segment honestly; if a 200 m run is marginal, present the longest segment that meets spec + the full curve, labeled. Honesty beats a doctored number in front of these judges.                  |
| R8 | <b>8-Jun deck over-claims</b> — showing target plots as if they were measured results.                                                                          | Med  | Low            | Any chart in the design deck without an "illustrative / target" label.                                                  | Every trajectory/drift/ellipse chart in the 8-Jun deck is labeled <b>"illustrative / target, not measured."</b> The deck is a <i>design proposal</i> , not a results report.                                                                                                                                                         | If asked, state plainly: "These are design targets; measured EuRoC + Gazebo numbers land by 11 Jun."                                                                                                                             |



## Part 2 — SME Q&A Answer Bank

One paragraph each. Answers are written to be said out loud, in our own words, under pressure.

**Q1. "Your protection level — protection level *relative to what datum*? You have no absolute reference indoors."** Correct, and that's the whole point — we don't claim an absolute PL. DP7's own accuracy metric is *segment-RPE* (KITTI-style relative pose error over a sliding window), so our integrity bound is built on the **same relative quantity**: it bounds the error growth of the VIO solution over a segment, not position against a global frame. It's the GNSS-RAIM analogue mapped onto the aiding source DP7 actually specifies. When we say "the bound covers the true position 95%+ of the time," "truth" is the sim ground-truth trajectory from Gazebo/PX4, and "covers" means the segment-relative error stays inside the bound. We're explicit that this is a *relative* integrity bound — the honest claim indoors, not a borrowed absolute one.

**Q2. "k = 6.74 — your bound is ~1.3 m and the true error is ~0.3 m. You've trivially over-met it. That's not integrity, that's slack."** Two separate bounds, on purpose. The **DAL-C bound** uses  $k_{\text{ffd}} = 6.74$ , derived from  $\lambda_{\text{md}} = 45.0$  via `scipy.stats` (we can show the `ppf / isf` call) for a fault-free-detection probability consistent with a DO-178C-as-if DAL-C allocation — that one is *deliberately* conservative; over-coverage is the safety margin, not a bug. The bound we *operate and tune on* is the **operational k = 2.45** ( $\approx 95\%$ ), which is the one that tracks the true error tightly and the one the breathing ellipse renders. We show both so the judge sees we understand the gap between an operating monitor and a certification-grade bound. The headline metric is the **NEES test**: across the run, the normalized estimation error squared stays inside its  $\chi^2(95\%)$  two-sided envelope, which is the statistical proof that the *operational* bound bounds — not slack, consistency.

**Q3. "Your observability index D is just a condition number. Dress it up however you like."** It's grounded in the actual VIO observability literature, not cosmetic. The right framing is the **OC-EKF** result (Hesch & Kottas, Huang) that VIO has four unobservable directions — global position (3) and yaw about gravity (1) — and that filter inconsistency comes from spuriously gaining information in those directions. D is the **whitened eigendecomposition of the local observability / information matrix**: it tells us *which state directions vision is currently informing and which it is not*, which is exactly the Zhang & Singh (ICRA 2016) degeneracy-direction idea. The condition number is the scalar summary; the *eigenvectors* are the payload — they say *that yaw/scale is going blind*, not merely *something is ill-conditioned*. For the live demo we run the scalar covariance-trace proxy; the full whitened decomposition is the offline/stretch artifact, presented as derived.

**Q4. "Gazebo is too easy — no motion blur, no rolling shutter, perfect synthetic features. Your VIO never sees the hard cases."** Agreed, which is exactly why our *guaranteed floor and our integrity demo both run on EuRoC*, not on Gazebo. EuRoC MAV is real flight data with real motion blur, real rolling-shutter artifacts, real IMU noise, and known-hard sequences (V2\_03, MH\_05). OpenVINS converging there is a meaningful result; converging in Gazebo is the *deliverable*, not the proof. The split is deliberate: Gazebo gives us the controllable kill-the-camera fault injection we can't safely stage on a real dataset, EuRoC gives us the realism. We're not betting the integrity claim on synthetic ease — the synthetic environment is the fault-injection rig, the real data is the credibility.

**Q5. "There's no calibration noise in sim. Real cam-IMU extrinsics and time offset are never that clean. Your whole filter assumes perfection."** In sim we *know* the extrinsics and the time offset exactly from the SDF and PX4 model, so we deliberately turn extrinsic/time-offset estimation **off** for the Gazebo run — we're not pretending to solve a problem the sim doesn't pose. The realism check is EuRoC, where we use the dataset's published **Kalibr** calibration and let OpenVINS run its normal online refinement. So the answer is two-part: sim isolates the integrity behavior with calibration controlled; EuRoC shows the same pipeline survives real, imperfect calibration. Calibration error is also exactly the kind of slow fault our integrity layer is *meant* to catch — it shows up as a persistent NEES bias, which is a feature demo, not a hidden assumption.

**Q6. "8 seconds of coasting on a tactical-grade IMU is trivial. An HG4930 barely drifts in 8 seconds. You're solving a non-problem."** Right — and that's *our headline number, stated honestly*, not a hidden weakness.  $\sim 0.4$  m of drift over an 8 s vision outage on an HG4930-class IMU (cross-checked to double-integrated bias/noise physics within  $\pm 20\%$ ) is precisely the point of the Honeywell framing: the IMU is the spine, vision is the aid that **bounds** the inertial drift, and a good IMU makes the coast cheap. The contribution isn't "we survive 8 s" — it's "we *know* the bound on that 8 s coast in real time, and we know within half a second when vision dropped out so we can trust-degrade gracefully." On a consumer IMU the same outage is  $\sim 6.3$  m and the integrity bound matters far more; we show all three tiers (consumer  $\sim 6.3$  m / industrial  $\sim 1.2$  m / HG4930  $\sim 0.4$  m) so the judge sees we understand *why* the IMU grade changes the integrity argument.

**Q7. "You just wired OpenVINS, evo, and a Streamlit dashboard together. Where's the engineering?"** The wiring is the *floor* — the four DP7 deliverables. The engineering is the **integrity layer that doesn't exist in OpenVINS**: a live, fault-hypothesis-derived relative integrity bound (the RAIM analogue for a vision aid), a NEES-validated consistency proof that the bound actually bounds, an observability index that tells us *which* state direction is going unobservable, and a continuous degradation manager R(D) that drives NOMINAL→DEGRADED→INERTIAL→RE-ACQUIRE off that bound. Gupta & Gao (ION *NAVIGATION*, "Integrity of Visual Navigation") frame this as an **open literature gap** — integrity monitoring for vision-based nav is not solved. We're not claiming to close it; we're building the student-scale, open-source instantiation of it on top of a solid VIO core, which is exactly the right way to not reinvent the MSCKF.

**Q8. "Is this even buildable by two second-years in four days?"** Yes, because we engineered the scope to *be* buildable, with a guaranteed floor and a hard line around what's stretch. By 11 Jun, high-confidence: OpenVINS out-of-box on EuRoC, OpenVINS on a Gazebo Harmonic tunnel, drift  $< 1.5\%$ , estimate-vs-GT plots, documented ROS2 workspace — that's **all four DP7 deliverables** with stock tools. The live integrity demo is medium-high confidence because it's **Python consuming what OpenVINS already publishes** — no C++ fork on the critical path. The full per-feature RAIM math is explicitly *stretch/offline*, shown as derived on slides, never bet on. The riskiest single item is Day-0 (Windows→Ubuntu), which is why it's the must-do with a cloud-GPU fallback. We descope *toward* a demo we can guarantee, not away from one.

**Q9. "Why MSCKF? Why not a modern optimization-based backend like VINS-Fusion or ORB-SLAM3?"** Three reasons, all deliberate. First, **integrity**: the MSCKF maintains an explicit state covariance every step, which is exactly the quantity our protection-level proxy and NEES test consume — an optimization backend gives you a MAP estimate but the marginal covariance is more work to extract live. Second, **bounded, real-time compute**: the MSCKF marginalizes features instead of keeping them in the state, so cost is bounded and deterministic, which matters for an on-board GPS-denied claim. Third, **fit to the Honeywell framing**: a filter is the natural home for "IMU is the spine, vision bounds the drift." We cite VINS-Fusion as our backup VIO and ORB-SLAM3 as the accuracy benchmark, and we use OpenVINS's `ov_eval` because it natively reports the NEES we need — the tool choice follows the integrity argument.

**Q10. "How fast do you actually detect a vision fault, and what triggers the degrade?"** Sub-0.5 second, by design and rehearsed. The trigger is twofold: tracked-feature count on the OpenVINS feature track collapsing toward zero, and the covariance trace on `/ov_msckf/odomimu` beginning to inflate. The degradation\_manager runs a continuous R(D) state machine — NOMINAL → DEGRADED → INERTIAL → RE-ACQUIRE — and crosses into INERTIAL within a few filter updates of the camera going black, which at VIO rates is well under half a second. The health\_cockpit flips the trust indicator to red on that same event, and the integrity ellipse begins to **bloom** as the bound grows to cover the now-coasting estimate. That "we know within half a second" is one of the three things we say on stage on purpose — fast, observable fault detection is the deliverable, not a side effect.

**Q11. "Your breathing ellipse is just covariance visualization. What makes it an *integrity* bound and not a pretty plot?"** The ellipse is the *rendering*; the integrity content is what sizes it. It's scaled by the protection-level proxy (covariance trace → k-scaled bound) and we **validate** that the rendered bound actually contains the truth using NEES against sim ground truth — so it's not just drawing  $1\sigma$ , it's drawing a bound we've statistically shown covers the true position at the stated confidence. When the camera dies and the ellipse blooms to  $\sim 1.0$  m chasing the moving truth dot, that bloom is the protection level growing exactly as the inertial-only uncertainty grows, and the truth dot staying inside it is the live evidence the bound is honest. A covariance plot shows you *uncertainty*; an integrity bound makes the testable claim "the truth is inside this, 95%+ of the time" — and we test it.

**Q12. "What if OpenVINS diverges live on stage instead of degrading gracefully?"** Then we drop to rung 2 of the safety ladder — `ros2 bag play` of a golden run recorded during the 11 Jun dry-run — and the integrity\_monitor consumes that bag *identically* to live, so the kill-camera beat (trust red < 0.5 s, ellipse bloom, recover) plays exactly the same. Rung 3 is a backup screen recording if even replay hitches. Crucially, the integrity layer is decoupled from VIO health by design: it's a separate Python node reading published topics, so a `vio_core` wobble doesn't take down the monitor or the cockpit. And because the demo can run on EuRoC independent of Gazebo, "VIO diverged in *our sim*" never becomes "the demo is dead" — it becomes "let me show you the same beat on real EuRoC data."

**Q13. "What's the single biggest thing that kills this project, and why aren't you worried about it?"** Day-0: getting off Windows onto a native Ubuntu 22.04 ROS2 Humble + Gazebo Harmonic stack. It's risk R1, ranked highest, because losing 8 Jun to environment hell compresses everything. We're not worried because we've made it the *first* thing we do, with a tested fallback chain: native dual-boot first (best GPU/timing), cloud GPU box second (Lambda/Paperspace, Ubuntu image, same day), and — for the integrity node specifically, which only needs a rosbag — WSL2 as a last resort since it doesn't need live Gazebo rendering. Every downstream risk has the EuRoC safety net underneath it, so even a bad sim day still produces all four DP7 deliverables plus the integrity demo on real data. The plan is designed so that the worst realistic outcome is "less impressive," never "nothing to show."

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## PART III — REFERENCE

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### 12. Honeywell alignment & future scope

**Sim-to-real:** the architecture transfers unchanged — swap sim stereo+IMU for a real rig; OpenVINS' online cam-IMU extrinsic + time-offset calibration handles mis-sync (demonstrated in §8.4); the integrity layer is sensor-agnostic. Because the sim IMU is parameterized to the **HG4930 CA51** (inside the **HGuide n580** family) and cross-checked to physics ( $\pm 20\%$ ), the bounded-error story is quantified against real Honeywell hardware; cite Honeywell's **HGuide-n580-with-ROS** whitepaper; name the one bench measurement that would validate the dead-reckoning envelope on silicon (a static HG4930 Allan-variance log) as the first Student-Research-Project experiment.

**Certification framing (substance, not decoration):** pick **DAL C**, justified from a stated hazard (loss of bounded indoor nav → controlled flight into terrain / asset loss in a warehouse = *major*, not *catastrophic* → DAL C → integrity risk  $1 \times 10^{-5}/\text{hr}$  →  $k_{\text{ffd}}=6.74$ ). Three modeled FMEA failure modes: camera blackout / feature starvation (D + feature-count gate → degrade); IMU bias jump (NEES/innovation → flag + inflate process noise); feature-association fault single + correlated (per-feature NIS + global frame NIS → exclude/degrade). Architect **as if** targeting DO-178C(C)/DO-254/ARP4754A/ARP4761 — modular independently-testable ROS2 nodes, documented failure modes, stated DAL — *not* a full requirements-traceability/MC/DC artifact (that's roadmap delta #3).

**"What would make *IB* a certifiable Protection Level" (seeds the Student Research Project):** 1. A **global PL via an absolute aid** (UWB / loop closure) — our *IB* is relative (the UWB clip is the proof-of-concept). 2. Characterize + over-bound the **non-Gaussian / heavy-tailed vision residual** with an empirically-validated CDF over-bound, validated against a real HG4930 Allan log. 3. Tighten to  $1 \times 10^{-7}/\text{hr}$  for catastrophic functions with formal DO-178C requirements coverage (traceability, structural coverage, MC/DC).

**Productization / HANA roadmap:** A.E.T.H.E.R is **one pluggable layer of a HANA-style stack** — the same node graph has labelled slots for **MagNav** · **terrain-aided** · **radar-velocity (HRVS)** · **LEO-satellite** aiding nodes. Literally Honeywell's layered HANA vision.

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### 18. Presentation — the deck & delivery

The deck is built as **"Flight-Test Report SV-026"** — seven numbered drawing sheets of a Honeywell navigation flight-test report, not a pitch deck. Drafting-paper off-white canvas, steel-blue technical linework, serif title-blocks with `DWG 0N/07` kickers, monospace for *every* machine string (ROS2 topics, k-values,  `$\lambda_{\text{md}}=45.0$` ), and **Honeywell red rationed to exactly two meanings:** the integrity bound / 1.5%-3.0 m threshold, and fault/blackout events. The recurring motif is the **dimensioned integrity ellipse over a truth dot**, which doubles as the top-right page-marker and visibly *breathes* across the sheets (tight+green → dual-ring → bloomed+red → collapsed).

18.1 The seven sheets

| # | Sheet                     | Hero visual                                                                                 | Headline                                              |
|---|---------------------------|---------------------------------------------------------------------------------------------|-------------------------------------------------------|
| 1 | Cover                     | tunnel-corridor est-vs-GT trajectory + the dimensioned ellipse                              | the spine, as the report abstract                     |
| 2 | Problem, reframed         | drift-%-vs-distance curve under the red 1.5% limit, PASS band shaded                        | "how wrong am I right now, and do I know it in time?" |
| 3 | Why Honeywell — the wedge | IMU-spine datum + vision-bounds-drift arm + the 3-grade IMU ghost ladder                    | "GNSS has RAIM. Vision aiding has none. We build it." |
| 4 | Architecture              | the five-node ROS2 schematic (integrity_monitor the only red node) + degradation ribbon     | "any subset runs — graceful by construction"          |
| 5 | The novelty               | the dual-bound dimensioned ellipse ( $k=2.45$ inside $k_{\text{ffd}}=6.74$ ) + bound ledger | "a live, fault-derived RELATIVE integrity bound"      |
| 6 | The demo                  | three time-stamped breathing ellipses chasing a moving truth dot + beat sheet + numbers     | "kill the camera, watch the bound breathe"            |
| 7 | Build plan                | the two-lane critical-path Gantt (red Day-0 risk, FLOOR locked after Day 2, dashed stretch) | "de-risked to a guaranteed floor"                     |

18.2 Delivery

- **The spine, said verbatim ×3** — open (sheet 1), demo (sheet 6), close (sheet 7): *"When the camera dies, our drift bound still covers the true position 95%+ of the time — and we know within half a second."*
- **3-minute path:** spine + reframe + journey [1–2] → Honeywell stakes + wedge [3] → architecture [4] → the bound, one number [5] → demo + the measured ablation, spine again [6] → roadmap close, spine a third time [7].
- **Q&A answer-bank:** see §16 — every likely SME kill-shot has a one-paragraph answer rehearsed.

16. Headline metrics table (the one slide SMEs photograph)

| Metric                                 | Target / spec                                                                                               | Honeywell analogue              |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------|---------------------------------|
| Translational drift                    | $< 1.5\% / 200\text{ m}$ ( $\leq 3.0\text{ m}$ terminal), KITTI segment-RPE, origin-aligned                 | bounded/known error             |
| Nominal ATE                            | sub-metre over hundreds of m (EuRoC-class)                                                                  | vision-aided accuracy           |
| IB coverage — operational ( $k=2.45$ ) | $\geq 95\%$ held-out                                                                                        | GNSS HPL/VPL alert (RAIM)       |
| IB coverage — DAL-C ( $k=6.74$ )       | $\sim 100\%$ by construction (exceedance $1.4 \times 10^{-10}$ )                                            | conservative integrity envelope |
| Fault-detection latency                | $< 1.0\text{ m} / < 10\text{ frames}$                                                                       | integrity / health monitoring   |
| Missed detections                      | 0                                                                                                           | fault detection & exclusion     |
| False-alarm rate                       | $< 1\%$ of nominal frames                                                                                   | continuity / availability       |
| Inertial-outage error                  | $\approx 0.4\text{ m @ } 8\text{ s}$ (HG4930); $\sim 1.2\text{ m}$ industrial; $\sim 6.3\text{ m}$ consumer | graceful degradation to INS     |
| Dead-reckoning vs physics              | within $\pm 20\%$ of analytic envelope                                                                      | bounded/known error, validated  |
| Transition spike                       | continuous $R(D) < \text{binary (measured)}$                                                                | no-jump deep fusion             |
| Trust metric                           | live 0–1, green/amber/red                                                                                   | INS analogue of RAIM            |

Appendix A — integrity-bound derivation (scipy-verified)

**A.1 MSCKF measurement, after null-space projection.** For feature  $i$ :  $\mathbf{r}_i = \mathbf{H}_{x,i} \tilde{\mathbf{x}} + \mathbf{H}_{f,i} \tilde{\mathbf{f}} + \mathbf{n}_i$ ,  $\mathbf{n}_i \sim N(0, \mathbf{R}_i)$ . Marginalize the feature by left-multiplying with  $\mathbf{A}^T$ , the orthonormal left null-space of  $\mathbf{H}_{f,i}$  ( $\mathbf{A}^T \mathbf{H}_{f,i} = 0$ ,  $\mathbf{A}^T \mathbf{A} = \mathbf{I}$ ):  $\mathbf{v}_i = \mathbf{A}^T \mathbf{r}_i = \tilde{\mathbf{H}}_{x,i} \tilde{\mathbf{x}} + \mathbf{A}^T \mathbf{n}_i$ , with  $\tilde{\mathbf{H}}_{x,i} = \mathbf{A}^T \mathbf{H}_{x,i}$ ,  $\tilde{\mathbf{R}}_i = \mathbf{A}^T \mathbf{R}_i \mathbf{A}$ ,  $\mathbf{S}_i = \tilde{\mathbf{H}}_{x,i} \mathbf{P} \tilde{\mathbf{H}}_{x,i}^T + \tilde{\mathbf{R}}_i$ .

**A.2 slope\_max — the worst-case undetected-bias RAIM slope.** Parameterize a fault  $\mathbf{b} \cdot \mathbf{u}$  along unit measurement direction  $\mathbf{u}$ . Position error  $\Delta \mathbf{p}_{xy}(u) = [\mathbf{K}_i]_{\{xy\}} \mathbf{A} \mathbf{u} \cdot \mathbf{b}$  ( $\mathbf{K}_i = \mathbf{P} \tilde{\mathbf{H}}_{x,i}^T \mathbf{S}_i^{-1}$ ); test non-centrality  $\lambda(u) = \mathbf{b}^2 \cdot (\mathbf{A} \mathbf{u})^T \mathbf{S}_i^{-1} (\mathbf{A} \mathbf{u})$ . The slope =  $\max_u \frac{\|[\mathbf{K}_i]_{\{xy\}} \mathbf{A} \mathbf{u}\|_2}{((\mathbf{A} \mathbf{u})^T \mathbf{S}_i^{-1} (\mathbf{A} \mathbf{u}))}$  — a generalized eigenvalue problem  $\lambda_{\max}(M, N)$  with  $M = \mathbf{A}^T \mathbf{K}_{xy}^T \mathbf{K}_{xy} \mathbf{A}$ ,  $N = \mathbf{A}^T \mathbf{S}_i^{-1} \mathbf{A}$ . The simplified  $\max \frac{\|[\mathbf{K}_i]_{\{xy\}}\|_2}{\|[\mathbf{K}_i]_{\{xy\}}\|_2}$  is a stated **conservative over-bound** (it can only over-state the bound — the safe direction).

**A.3 Unit bookkeeping.**  $\mathbf{A}$  orthonormal  $\Rightarrow \mathbf{A}^T$  preserves pixel units;  $\text{slope\_max}$  is in m/px,  $\text{p\_bias}$  in px, product in metres.

**A.4 p\_bias from the non-central  $\chi^2$ .**  $\text{dof}=2$ ,  $\text{P\_FA}=1 \times 10^{-3} \rightarrow \text{T\_D} = \text{chi2.ppf}(0.999, 2) = 13.816$ .  $\text{P\_MD}=1 \times 10^{-3} \rightarrow \text{solve } \text{ncx2.sf}(13.816, 2, \lambda_{\text{md}}) = 0.999 \rightarrow \lambda_{\text{md}} = 45.0$ ,  $\text{p\_bias} = \sqrt{45.0} \cdot \sigma_{\text{meas}} = 6.71 \cdot \sigma_{\text{meas}}$ .

**A.5 The dual Protection Level.**  $\text{IB\_assoc}(k) = k \cdot \sqrt{\lambda_{\max}(\mathbf{P}_{xy})} + \text{slope\_max} \cdot \text{p\_bias}$ . Operational  $k = \sqrt{(\text{chi2.ppf}(0.95, 2))} = 2.45$ . DAL-C  $k_{\text{ffd}} = \sqrt{(\text{chi2.ppf}(1 - 1.39 \times 10^{-10}, 2))} = 6.74$  (per-sample risk from  $\text{P\_HMI}=1 \times 10^{-5}/\text{hr}$  at 20 Hz; exceedance  $\text{chi2.sf}(6.74^2, 2) =$

$1.39 \times 10^{-10}$  ).

**A.6 Observability index**  $\mathbf{D}$  . Whiten ed  $\tilde{\Lambda}_v = P^{1/2} \Lambda_v P^{1/2}$  (dimensionless info-ratio eigenvalues); flag eigenvectors below  $0.10 \times$  the per-axis healthy spectral floor; online-classify the motion regime (const-vel / pure-rot / planar), form the analytic degenerate basis, exclude the 4 OC-EKF nulls;  $\mathbf{D}$  = normalized flagged-energy on that subspace.

**A.7 Worked single-feature example.**  $\sigma_{\text{meas}}=1.5$  px ,  $\text{dof}=2$  :  $p_{\text{bias}} = 6.71 \cdot 1.5 \approx 10.1$  px ;  $\text{slope}_{\text{max}} \approx 0.05$  m/px  $\rightarrow$  worst undetected position contribution  $\approx 0.50$  m;  $\sigma_{\text{pos}} = \sqrt{\lambda_{\text{max}}(P_{xy})} \approx 0.12$  m ,  $k_{\text{ffd}}=6.74 \rightarrow$  fault-free term  $\approx 0.81$  m;  $\mathbf{IB}_{\text{assoc}}(\text{DAL-C}) \approx 1.31$  m (operational  $k=2.45 \rightarrow \approx 0.79$  m). Reconciliation: actual instantaneous error  $\approx 0.3$  m;  $\mathbf{IB}$  is the trust *envelope* (worst-case-undetected, relative datum), not consumed drift budget; accumulated 200 m drift stays  $< 3.0$  m.

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## The one-line close (ends on the spine)

"DP7 is fundamentally an Assured-PNT problem. We built integrity-aware, observability-conditioned Visual-Inertial Odometry that knows when it can be trusted, expresses that trust as a fault-hypothesis-derived relative integrity bound — the RAIM analogue for the vision aid, on the exact segment-RPE quantity DP7 specifies — and degrades gracefully to inertial dead-reckoning whose error we measured at  $\approx 0.4$  m per 8 s against a Honeywell HG4930-class IMU, matching first-principles physics to  $\pm 20\%$ . When the camera dies, our drift bound still covers the true position 95%+ of the time — and we know within half a second — the same resilient-navigation philosophy HANA brings to contested airspace, scaled to an indoor drone and built entirely on open source."

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Companion file: the exhaustive iteration master plan is at `D:\Work\project-pramana\DP7-A.E.T.H.E.R-winning-pPlan.md` . This bible is the clean canonical reference.